Universal Plastic Shredder V2.0

Electrical Control & Power System

Portland State University
Maseeh College of Engineering & Computer Science
Department of Electrical & Computer Engineering

Final Report — Rough Draft

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Contents

1 Executive Summary

- Ichor Systems generates significant plastic waste (PVC, polypropylene, LDPE) during semiconductor equipment manufacturing, currently sent to landfill.
- Commercial industrial shredders cost $10,000+ and consumer units cannot handle the thick rigid sheets (0.25–0.5 in).
- Our project designs the electrical control and power system for a custom industrial-grade plastic shredder, capable of shredding into 1-inch pieces.
- Key elements: VFD-controlled 3 HP motor at 70–90 RPM output, two-hand deadman control, hardwired E-stop, jam-detection auto-recovery with 3-strike lockout, NEMA-enclosed PLC and HMI.
- Total budget: $3,000 (EE portion approximately $1,000).

Full-load shredder validation is deferred — see Section 10.2 (Load Testing). The ME team's shredder head is not functional at the time of this report, so the executive summary records signal-chain and bench validation results only; loaded-throughput numbers are out of scope until the mechanical build is delivered.

2 Introduction

2.1 Background and Motivation

- Ichor Systems is a Portland-area semiconductor equipment manufacturer.
- Their manufacturing processes produce substantial volumes of rigid plastic sheet waste.
- Existing disposal route: landfill – environmentally and economically suboptimal.
- A previous 2023–2024 PSU capstone team built a 2 HP household shredder for the EPL. This project scales to industrial-grade.

2.2 Project Scope

- EE team responsibility: motor control, safety circuits, PLC logic, HMI, power distribution.
- ME team (separate) handles mechanical frame, blade assembly, gearbox.

- Integration plan: mechanical structure delivered with motor mount; EE team commissions controls.

Plastic waste from semiconductor equipment manufacturing has historically been treated as a single-use disposal stream: thick rigid PVC, polypropylene, and LDPE offcuts from machine bring-up and rework are heavy, bulky, and expensive to haul out. Reducing those off-cuts to a size and form that can be re-fed into compatible polymer streams turns a disposal liability into a recycling input, which is the operating premise of the broader *circular economy* model for industrial plastics — keep material in use at its highest economic value for as long as possible, and design recovery paths that minimize new resin demand. A shop-floor shredder sized for Ichor's actual offcut stock (0.25–0.5 in rigid sheets) is the upstream step that makes the rest of that recovery loop viable; without it, the material is too dimensionally awkward to feed downstream sorting, washing, or extrusion equipment. This project's contribution is the electrical and control side of that shredder — specifically the motor-control, safety, and operator-interface subsystems that make a 2–5 HP industrial cutter usable and safe in a manufacturing environment.

3 Requirements

3.1 Must-Have

- Shred 0.25–0.5 in plastic sheet into approximately 1-inch pieces.
- Controllable VFD supporting 3–5 HP motor at 70–90 RPM output shaft speed.
- Follow NFPA 70 (NEC) electrical design guidance.
- Include emergency stop, overcurrent/overload protection, at least one safety interlock.
- Include an HMI for operator control and status display.
- Stay within $3,000 total project budget.

3.2 Should-Have

- Clear status and fault indicator lights.
- Follow NEMA enclosure guidelines.
- Clearly labeled controls.

3.3 May-Have (if time/budget allow)

- Wireless monitoring (read-only).
- Data logging of operating hours and faults.
- Additional safety sensing (capacitive touch, thermal).

4 System Architecture

4.1 Hardware Subsystems

- **AC power input:** 120 VAC mains → main disconnect → branch breakers → VFD and 24V PSU.
- **VFD:** Huanyang HY02D211B-T (2.2 kW, 110V single-phase input, 220V 3-phase output boost-type). Controlled via FOR/REV/RST digital inputs.
- **Motor:** the production motor is an **IronHorse MTCP2-002** (2 HP, 230 V low-voltage, FLA 5.93 A at 60 Hz, 4-pole, full-load speed 1735 RPM, NEMA 145TC frame, TEFC IP55, VFD-rated 1.0 SF, 10:1 CT / 20:1 VT adjustable-speed range). Initial bench/test used a GE 5KE182BC205B (3 HP, 230 V, 1750 RPM, FLA 7.6 A); the VFD parameter set in the panel today still reflects the bench values pending reprogram with the IronHorse nameplate (see Appendix 16.4). Full nameplate is in `VFD/Motor_Nameplate.md` in the repository.
- **Brake resistor:** 75 Ω, 500 W, connected to VFD's P/Pr terminals to handle regenerative energy during deceleration and reversal.
- **PLC:** AutomationDirect H2-DM1E (Do-more CPU) on DL205 base, D2-08ND3 discrete input module.
- **HMI:** AutomationDirect C-more Micro-Graphic EA1-T4CL 4-inch color touch panel, connected to the H2-DM1E CPU serial communication port (operator commands to the PLC; status/data back to the panel). Powered from the shared Mean Well NDR-480-24 24 VDC supply.
- **Safety circuit:** Hardwired E-stop loop and two-hand deadman start; motor contactor energized through the safety chain. Access guarding at the feed is handled by the ME team's mechanical design, so no separate electrical lid interlock is used.
- **24V DC supply:** Mean Well NDR-480-24 (480 W) for control voltage.

4.2 Signal Flow

- Operator inputs (deadman buttons, E-stop, reset) → PLC discrete inputs (D2-08ND3, sourcing mode).
- PLC outputs (D2-08TR relay module, common to PSU –V) → VFD FOR (Y7), REV (Y6), RST (Y5) inputs in sinking (NPN) mode.
- VFD fault relay (FC + FA, NO active-high) → PLC X5 input. X5 = HIGH when VFD trips on fault, LOW when healthy.
- HMI ↔ PLC via serial (H2-DM1E CPU serial COM port; matched baud, parity, and stop bits on both ends).

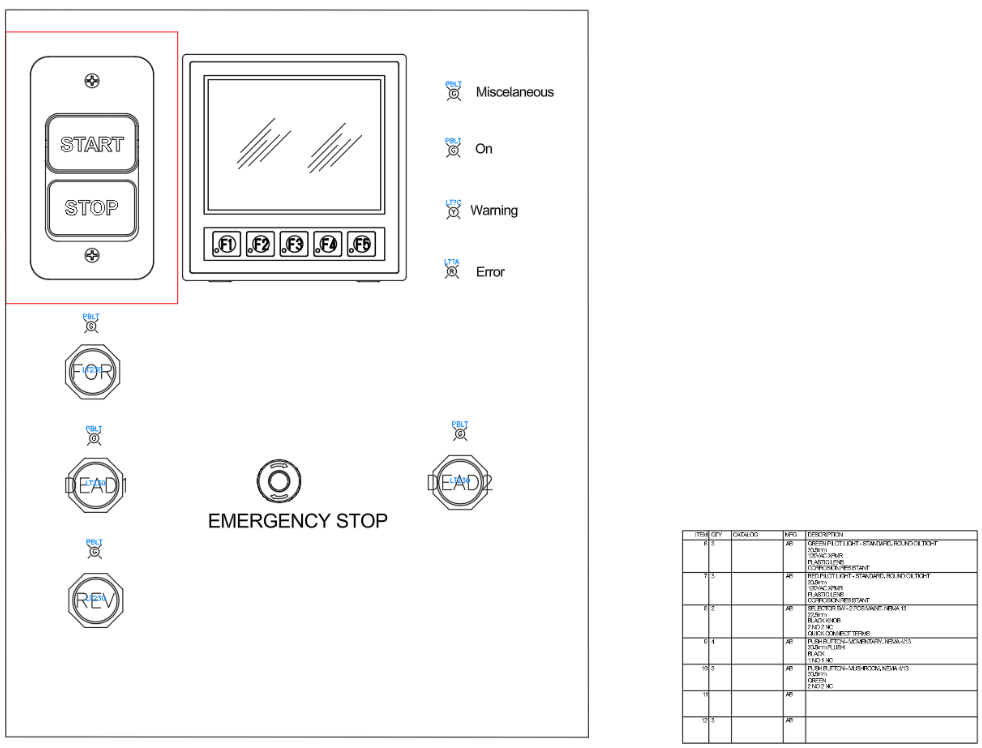

Figure 1: Operator front-panel CAD layout: HMI, power switch, forward / reverse and two deadman pushbuttons, emergency stop, and status indicators (Miscellaneous / On / Warning / Error). The START / STOP cutout (red highlight) carries the panel power switch.

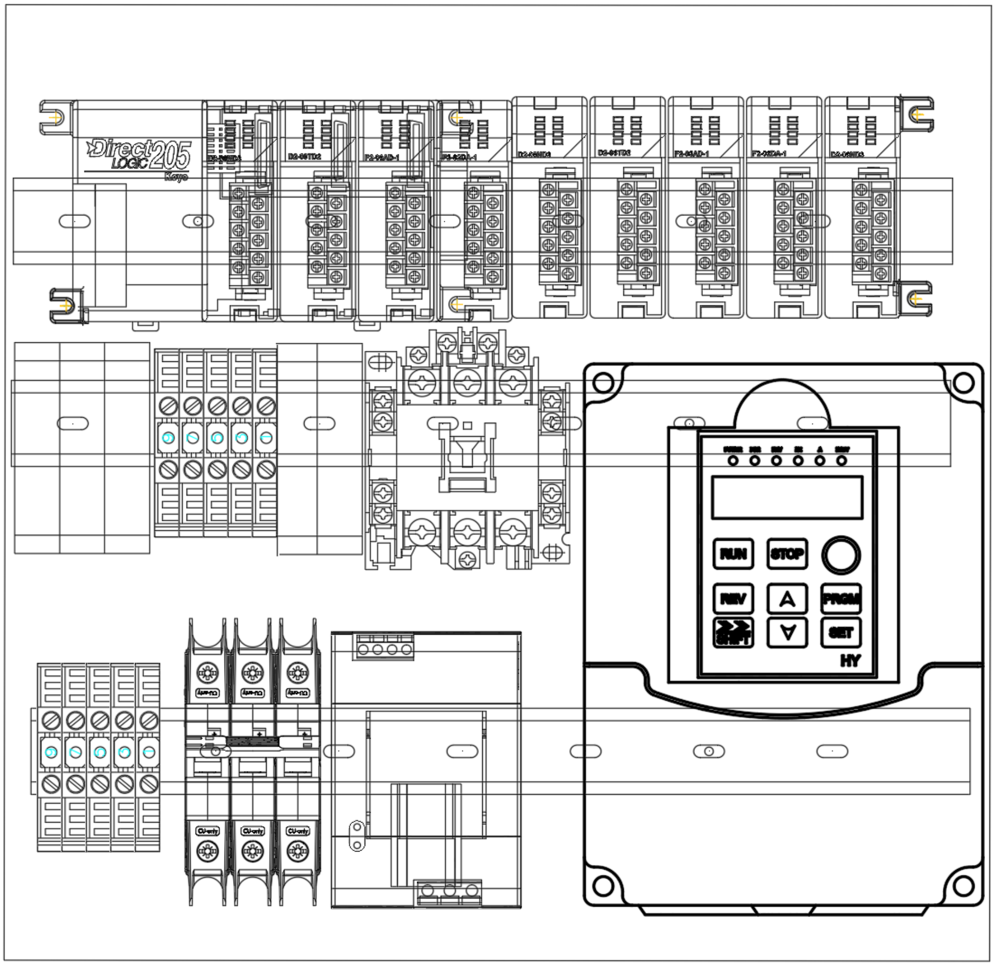

Figure 2: Internal panel layout: DirectLOGIC 205 PLC rack (top), control-wiring DIN-rail terminal blocks, the SD-N35 contactor branch, the 24 VDC supply, and the Huanyang HY02D211B-T VFD keypad on the right.

Figure 1: Control system block diagram — signal flow and functional domains.

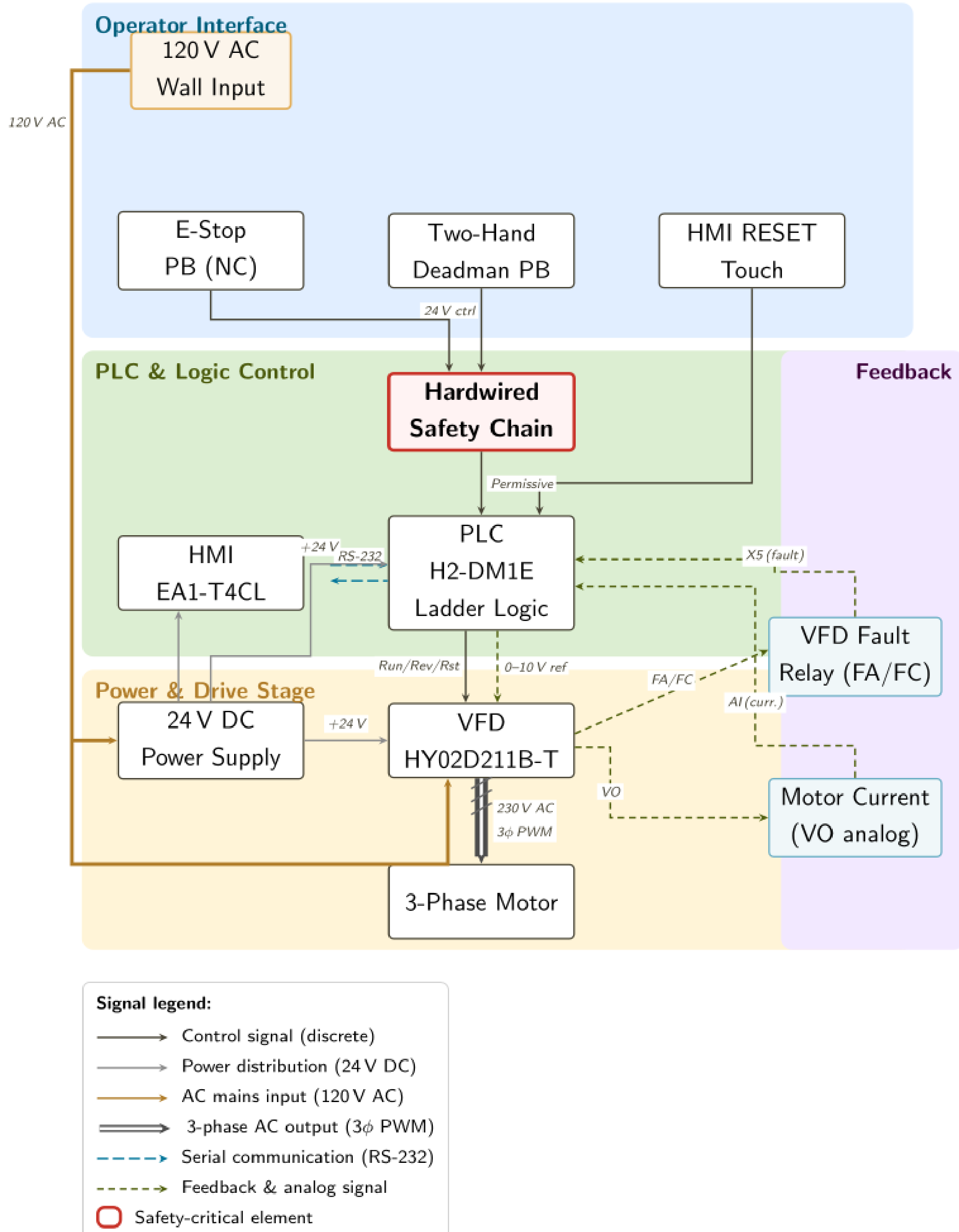

Figure 3: System block diagram — power flow (AC service → disconnect → contactor → VFD → motor) and signal flow (HMI ↔ PLC ↔ VFD, with the analog motor-current feedback path and the hardwired E-stop / deadman safety chain).

5 Hardware Design

5.1 Power Distribution

- Main disconnect / power switch: PowerTec 71008 two-pole (DPST) START/STOP switch, 120–230 VAC single-phase, rated 20 A / 3 HP max; switches both the line and neutral conductors.
- VFD branch breaker: 30 A C-curve DIN-rail (handles inrush, sized per NEC 430.52).
- 24V PSU branch breaker: 10 A C-curve DIN-rail.
- **Conductors:** power runs use 12 AWG copper (THHN/MTW), rated 20 A / 25 A / 30 A at the 60/75/90 °C ampacity columns of NEC Table 310.16 – well above the motor's 7.6 A FLA and the VFD input current. Control wiring uses 16–18 AWG, landed directly as straight stripped conductors at the terminal blocks (no ferrules used in this build).
- **Fuse / OCPD sizing:** branch and component fuses are sized at **125% of the protected component's rated continuous load** per NEC 210.20(A) / 215.3. (The motor branch breaker is the exception – NEC 430.52 permits a higher multiple of FLC to ride through inrush; running overload is handled by the VFD's electronic motor-overload function.)
- All branches fused individually at terminal blocks for downstream loads.

5.2 Motor and VFD

The Huanyang HY02D211B-T was selected over alternatives via a decision matrix (price, documentation, reliability). A brake resistor on the P/Pr terminals absorbs the regenerative energy from rapid reversal during the jam-clear cycle. The drive is programmed as shown in Table 1.

Table 1: Programmed VFD parameters (Huanyang HY02D211B-T).

Param	What it does	Value	Why this value
`PD141`	Motor rated voltage	230 V	Matches the GE test-motor nameplate (230 V low-voltage connection); sets the base of the V/Hz curve.
`PD142`	Motor rated current (FLA)	7.6 A	Nameplate full-load amps; sets the drive's current limit and electronic-overload reference.
`PD143`	Number of motor poles	4	Nameplate; 4-pole = 1800 RPM synchronous, so the drive resolves speed and slip correctly.
`PD144`	Motor rated speed	1750 RPM	Nameplate full-load speed; with the pole count it lets the drive estimate slip.
`PD002`	Run / frequency source	1 (external)	Speed comes from the external 0–10 V analog input (VI) driven by the PLC, not the keypad.
`PD052`	Relay output (FA/FC) function	02 (fault)	Closes the FA–FC relay on any fault so the PLC reads a jam/trip on input X5.
`PD054`	Analog output (VO) function	1 (current)	Outputs 0–10 V proportional to motor current to the PLC analog input for monitoring.
`PD123`	Over-torque action	3 (trip)	Over-torque self-trips the drive (which fires the fault relay); see the note below.

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Param	What it does	Value	Why this value
`PD124`	Over-torque threshold	150%	Set between no-load current (∼0.5 A) and measured jam current (>11 A) so normal shredding does not trip but a jam does.
`PD125`	Over-torque detection time	3.0 s	The over-torque must persist 3 s before tripping, so brief load spikes (feeding a piece) do not nuisance-trip.

A firmware-specific limitation surfaced during commissioning: the over-torque relay function (`PD052 = 12`) does not actually fire the FA/FC relay on the Huanyang revision we received when `PD123 = 2` (continue running). The working configuration used in this build is `PD052 = 02` (general fault indication) paired with `PD123 = 3` (over-torque triggers a self-trip, which then fires the fault relay) — see Table 1. The PLC then runs the unjam routine including a RST pulse to clear the VFD's latched fault state between retry attempts.

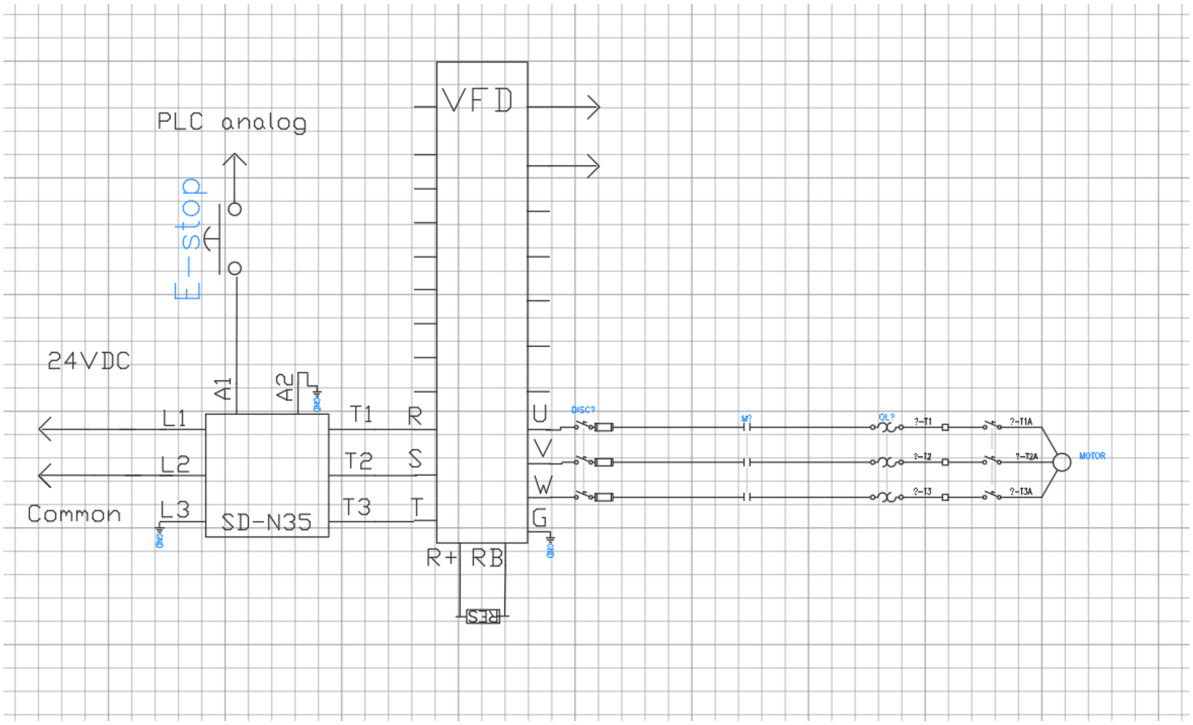

Figure 4: Power-circuit schematic. AC service (L1 / L2 / Common) lands on the SD-N35 contactor pole; the contactor coil A1 sits in the hardwired E-stop chain (24 VDC supply, top-left). Contactor output T1 / T2 / T3 feeds the VFD power inputs R / S / T; VFD outputs U / V / W drive the motor through the panel-mount disconnect, the overload, and the per-phase branches. The brake resistor lands on the VFD R+/RB terminals.

5.3 Safety Circuit

- E-stop: NC contact (E22B1) hardwired in series with the SD-N35 contactor coil. Pressing it drops the coil and opens the contactor, removing motor power independently of the PLC.
- Two-hand deadman: both buttons must be held continuously to enable the motor (evaluated by the PLC; see the Safety Functions assessment below).
- Access guarding: handled by the ME team's mechanical design (feed geometry), so no separate electrical lid interlock is used (this was a may-have, not a requirement).
- Motor contactor: the SD-N35 is energized through the hardwired safety chain, and the VFD is powered through it, so the safety chain has hardware precedence over the PLC.

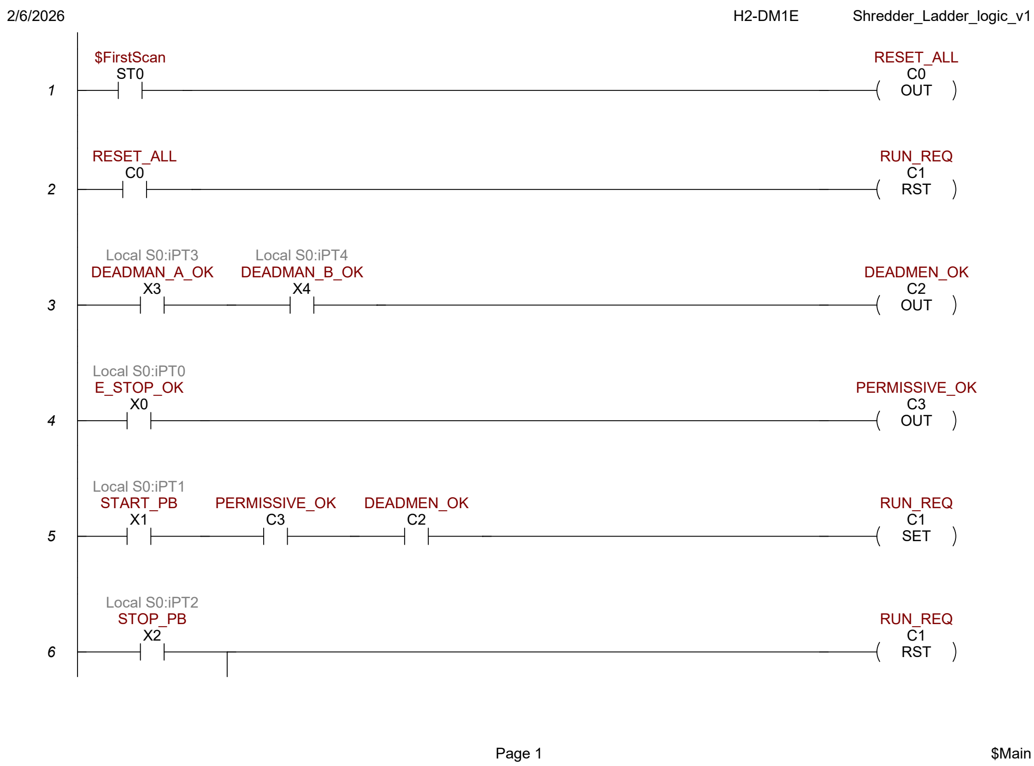

Figure 5: Hardwired safety circuit (E-stop and two-hand deadman). The Eaton E22B1 mushroom E-stop NC contact and both Autonics deadman pushbuttons sit in series with the SD-N35 contactor coil A1; pressing E-stop or releasing either deadman drops the coil, which opens the contactor pole feeding the VFD and removes motor power independently of the PLC. PLC X4 taps the coil node so the ladder can observe the safety state but cannot defeat it. Wiring complies with NFPA 79 §9.2 (hard-wired stop circuit independent of the programmable controller).

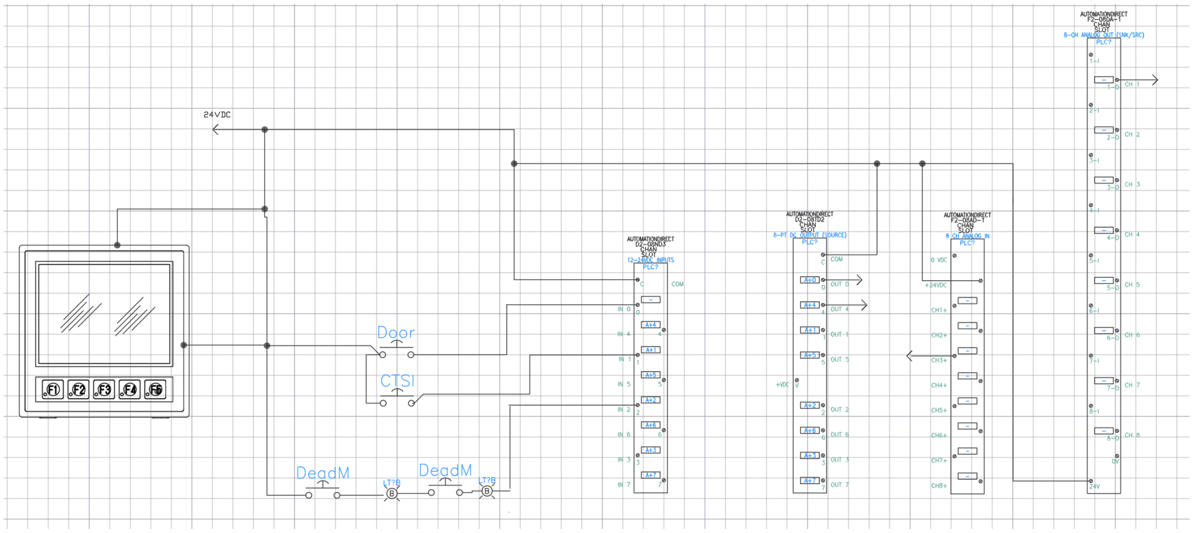

Figure 6: Control-side wiring: 24 VDC distribution from the Mean Well NDR-480-24, fused branches to the HMI, the two deadman pushbuttons, the door / interlock taps, and the PLC discrete-input terminal blocks. The HMI serial link (left) lands on the H2-DM1E CPU COM port; the +24 V / 0 V rails are shared across the PLC module power and the pushbutton commons.

5.4 Jam Detection Signal Chain

- VFD's multi-function relay (terminals FA, FB, FC) programmed for Fault Indication (`PD052 = 02`). **Only FA and FC are used** for jam detection; the FB (normally-closed) contact is left unconnected.
- Wired as an NO active-high pair: +24V (through a 500 mA fast-blow fuse) feeds the FA contact; the FC common returns to PLC X5 input on D2-08ND3 (Slot 2). When the VFD trips, FA–FC closes and +24V reaches X5.
- PLC sees X5 = LOW during normal operation, X5 = HIGH when VFD trips on any fault (over-torque, overvoltage, overheat, etc.).
- PLC ladder reads `[ X5 ]` on rising edge to detect new fault events.
- Wire-break failure mode is handled at the safety-circuit level by the independent hard-wired E-stop loop and motor contactor, rather than at the jam-signal level.
- Full terminal-by-terminal wiring is given in the I/O point list (Appendix B) and the VFD parameter/wiring table (Appendix D).

6 Software Design

6.1 PLC Ladder Logic Architecture

The control program runs on the H2-DM1E CPU in the DirectLOGIC 205 D2-09B-1 rack and was written in Do-more Designer. The program is 31 rungs long and is exported in `electrical/Circuit_Logic.pdf`. It implements a state machine for normal operation plus a 3-strike jam recovery sequence and a hard-lockout path.

Key Terms

Tag	Meaning		Tag	Meaning
FORWARDS (X0)	Forward pushbutton		STOP_MOTOR (C15)	Hard stop / lockout flag
REVERSE (X1)	Reverse pushbutton		PAUSE_MOTOR (C17)	Post-reverse coast-to-stop
DEADMAN_A_OK (X2)	Deadman button A		ERROR_RESET (C18)	Fault reset coil
DEADMAN_B_OK (X3)	Deadman button B		JAMMED_MOTOR (C20)	3-strike lockout flag
STOP_PB (X4)	E-stop NC contact monitor		CTSI_TRIG (C21)	Cap-touch trigger flag
OVERCURRENT (X5)	VFD fault relay input		CHANGE_SCREEN (C22)	HMI screen-change trigger
CAP_TOUCH_TRIP (X7)	Cap-touch safety input		TESTMAN_A_OK (C23)	HMI soft deadman A
FWD_LED (Y0)	Forward indicator LED		TESTMAN_B_OK (C24)	HMI soft deadman B
REVERSE_LED (Y1)	Reverse indicator LED		OVERCURRENT_TEST (C25)	HMI soft overcurrent test
DMAN_A_LED (Y2)	Deadman A indicator LED		E_STOP_TRIG (C26)	E-stop triggered flag
DMAN_B_LED (Y3)	Deadman B indicator LED		T0–T1	Overcurrent / unjam off-delay (2 s)

continued on next page . . .

Tag	Meaning		Tag	Meaning
E_LIGHT (Y4)	Error / fault indicator LED		T2	Lockout accumulating tmr (30 s)
VFD_RESET (Y5)	VFD fault reset output		T3	Post-reverse pause (2 s)
VFD_RUN_FWD (Y6)	VFD forward run output		T4	HMI screen timeout (3 s)
VFD_RUN_RVS (Y7)	VFD reverse run output		CT0	Jam-event counter (preset 3)
DEADMEN_OK (C0)	Both deadman buttons held		V100	HMI screen selector
HMI_STP_REQ (C1)	HMI stop button request		V2000	Speed setpoint (0–1600)
PERMISSIVE_OK (C3)	Run permissive latch		V3000	Scaled motor current (0–140)
FORWARDS_SET (C8)	Forward direction latch		WX0 / WY0	Analog in / out words
REVERSE_SET (C9)	Reverse direction latch		$FirstScan, $On	Do-more system bits
JAM_LATCH (C10)	Overcurrent jam latch			
RUN_REQUEST (C12)	Motor run command latch			
UNJAM_R (C13)	Unjam reverse stroke latch			

Inputs (X-registers) come from physical pushbuttons, the VFD fault relay, and a capacitive-touch safety sensor. C-registers hold latched internal states. Y outputs drive panel LEDs and the VFD's run / reset terminals. The full address-by-address reference — with electrical types, source modules, and detailed semantics — is reproduced in Appendix E.

6.1.1 State Machine

- **IDLE** → RUNNING when both deadman buttons held AND not locked out AND startup mask done.

- **RUNNING** → JAM_DETECTED on rising edge of X5 (VFD trip → fault relay fires → X5 goes HIGH).
- **JAM_DETECTED** → RST_PULSE if jam count < 3, → LOCKOUT if jam count ≥ 3.
- **RST_PULSE** (500 ms) → REVERSE (2 s).
- **REVERSE** → RESUME (brief delay) → back to RUNNING.
- **LOCKOUT** → IDLE only via dedicated reset button.

6.1.2 Jam Counter Reset Logic

- Counter (CT0) increments on each X5 rising edge (each new VFD fault triggers a jam event).
- Resets when motor runs cleanly for 30 continuous seconds (T2 accumulating timer).
- Also resets via the HMI Screen 3 ERROR_OFF button (ERROR_RESET, C18).
- Whichever condition fires first wins.

6.1.3 Startup Mask

- For first scan after PLC power-up, the FORWARDS_SET / REVERSE_SET pair is initialized via rung 4 (defaults to forward direction).
- In the active-high NO X5 wiring, no boot-time mask is required because the relay is naturally open at startup.

6.1.4 Rung-by-Rung Description with Ladder Logic Export

The 31 rungs are reproduced below directly from the Do-more Designer export (`electrical/Circuit_Logic.pdf`), with the actual ladder image on each page paired with a description of the rungs it contains. This lets a reviewer see the rung graphic next to the variable map (above) and the narrative below.

Page 1 of 7 — Rungs 1–5: Power-up Init and Deadman Permissive

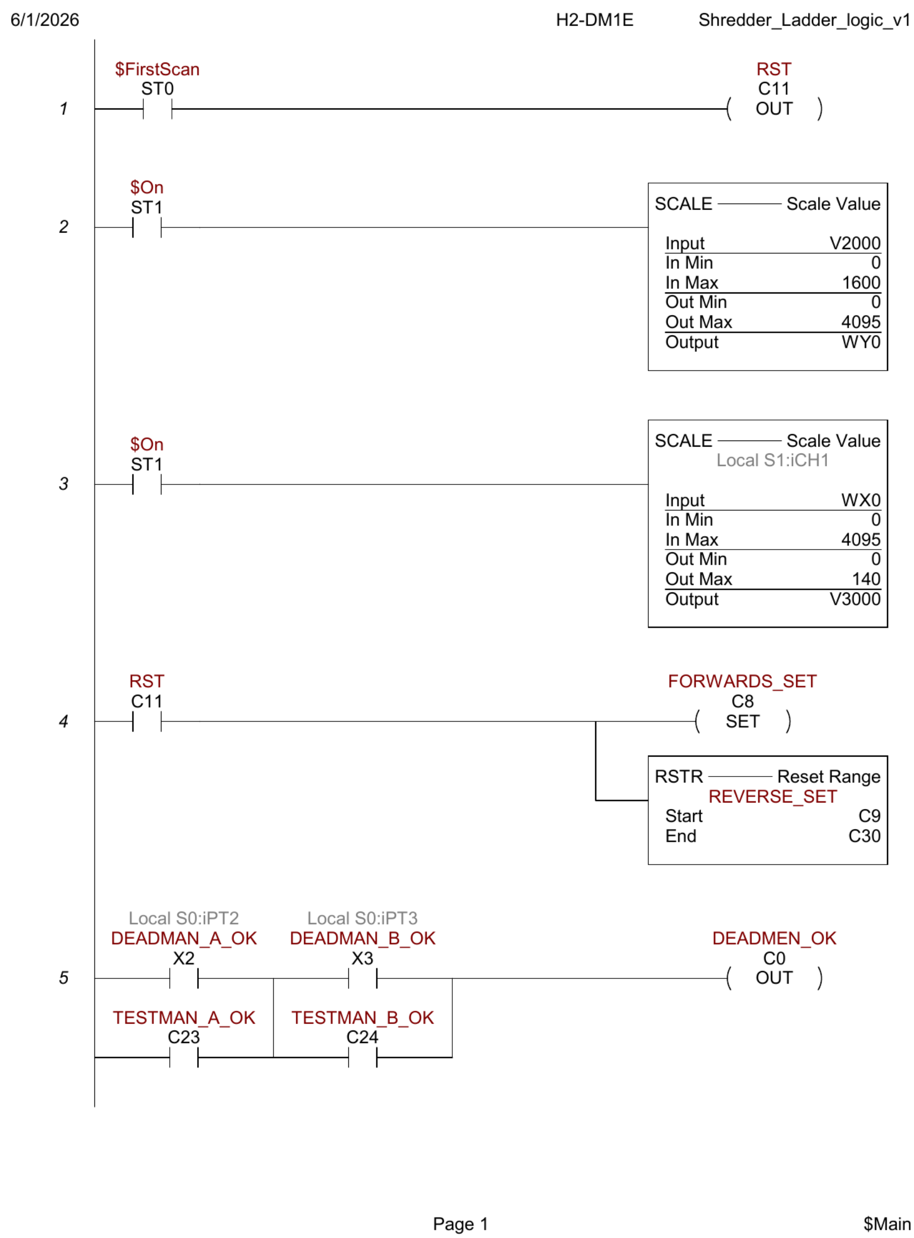

Figure 7: Ladder export page 1. Power-up init and deadman permissive (rungs 1–5).

- **Rung 1** ($FirstScan → RST C11): on the very first scan after power-up, C11 is reset, triggering the init path in rung 4.
- **Rung 2** ($On → SCALE V2000 [0–1600] → WY0 [0–4095]): every scan, the HMI speed setpoint V2000 is linearly scaled to raw DAC counts for the F2-08DA-2 CH1 → VFD VI.
- **Rung 3** ($On → SCALE WX0 [0–4095] → V3000 [0–140]): raw ADC from F2-08AD-1 CH1 (VFD VO motor current) is scaled to a 0–140 display value stored in V3000 (CURRENT tag).
- **Rung 4** (NOT C11 → SET FORWARDS_SET + RSTR C9–C30): initializes FORWARDS_SET and clears C9–C30 on first power-cycle scan.
- **Rung 5** ((X2 OR C23) AND (X3 OR C24) → OUT C0): computes DEADMEN_OK by AND-ing the two deadman paths (physical OR HMI-soft test buttons).

Page 2 of 7 — Rungs 6–10: RUN_REQUEST Gating and Run Outputs

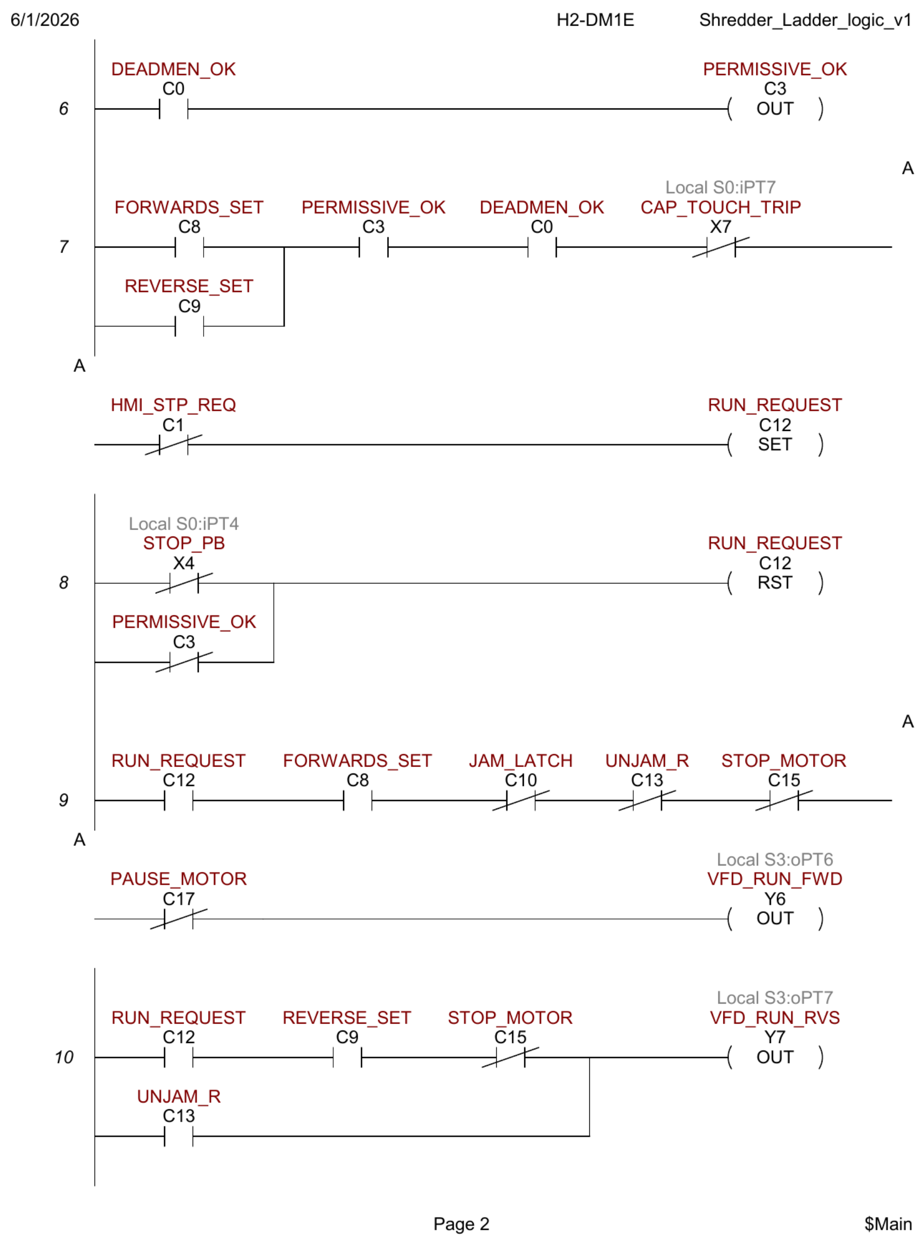

Figure 8: Ladder export page 2. RUN_REQUEST gating and run outputs Y6/Y7 (rungs 6–10).

- **Rung 6** (C0 → OUT C3): mirrors DEADMEN_OK to PERMISSIVE_OK so the permissive logic is referenced separately from the deadman coil.
- **Rung 7** ((C8 OR C9) AND C3 AND C0 AND NOT X7 → SET C12): RUN_REQUEST is set when a direction is selected, permissives are OK, and the capacitive touch is not tripped. The HMI STOP button (NOT C1) is a separate inhibit path.
- **Rung 8** (NOT X4 OR NOT C3 → RST C12): RUN_REQUEST is immediately reset if the E-stop opens (X4 LOW) or the deadman permissive drops.
- **Rung 9** (C12 AND C8 AND NOT C10 AND NOT C13 AND NOT C15 AND NOT C17 → OUT Y6): forward run output. Requires RUN_REQUEST + FORWARDS_SET + no active jam + not in reverse unjam stroke + no hard stop + post-reverse buffer elapsed.
- **Rung 10** ((C12 AND C9 AND NOT C15) OR C13 → OUT Y7): reverse run has two paths: manual reverse (RUN_REQUEST AND REVERSE_SET AND NOT STOP_MOTOR), or unconditional unjam reverse (UNJAM_R alone).

Page 3 of 7 — Rungs 11–16: Indicator LEDs and Direction Latches

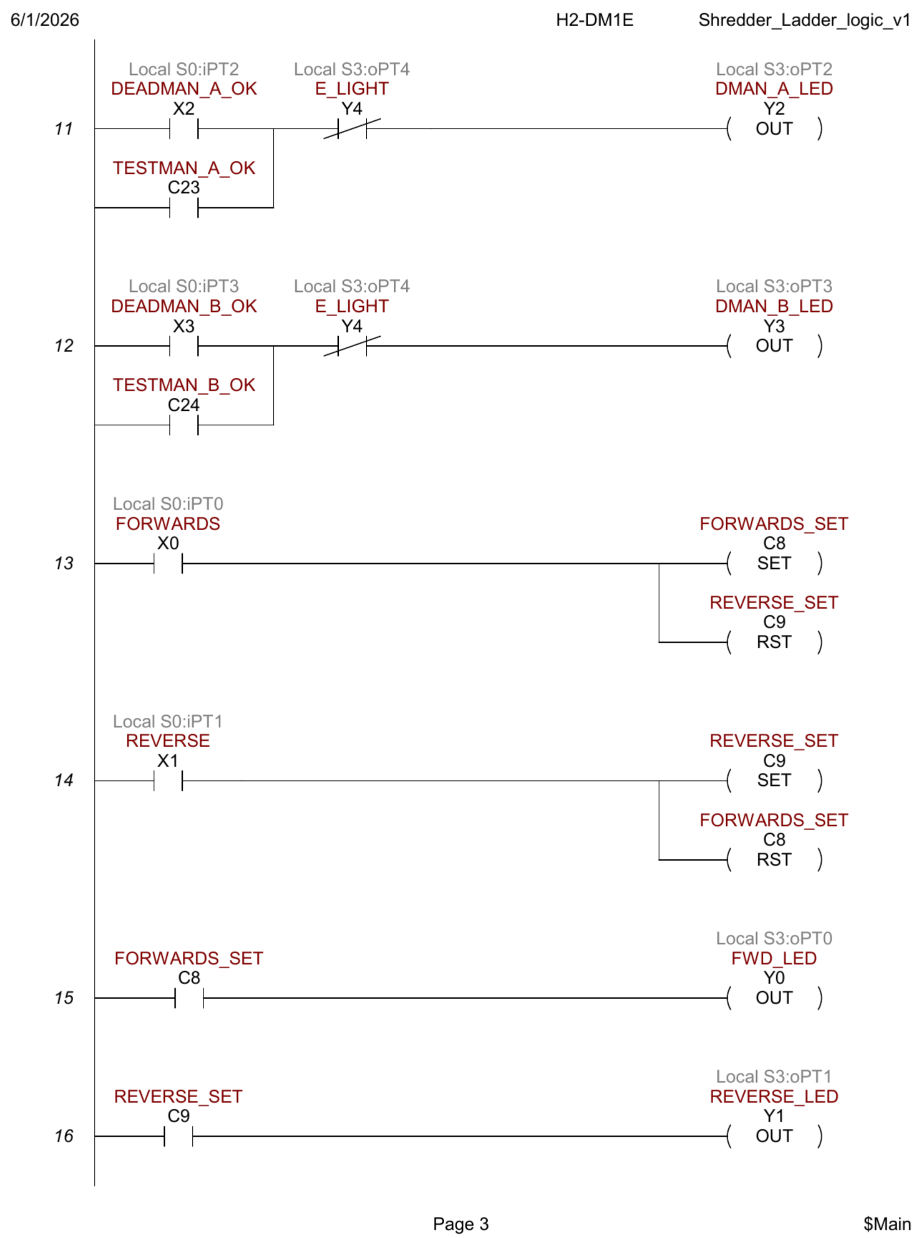

Figure 9: Ladder export page 3. Indicator LEDs and direction latches (rungs 11–16).

- **Rung 11** ((X2 OR C23) AND NOT Y4 → OUT Y2): deadman A LED lights when the button is pressed but only if the fault indicator (Y4) is not lit.
- **Rung 12** ((X3 OR C24) AND NOT Y4 → OUT Y3): identical pattern for deadman B.
- **Rung 13** (X0 → SET C8 + RST C9): forward pushbutton latches FORWARDS_SET and clears REVERSE_SET (mutual exclusion).
- **Rung 14** (X1 → SET C9 + RST C8): reverse pushbutton is the mirror image.
- **Rung 15** (C8 → OUT Y0): forward indicator LED ON whenever FORWARDS_SET is latched.
- **Rung 16** (C9 → OUT Y1): reverse indicator LED ON whenever REVERSE_SET is latched.

Page 4 of 7 — Rungs 17–21: E-stop Response, Jam Detection, and Unjam Stroke

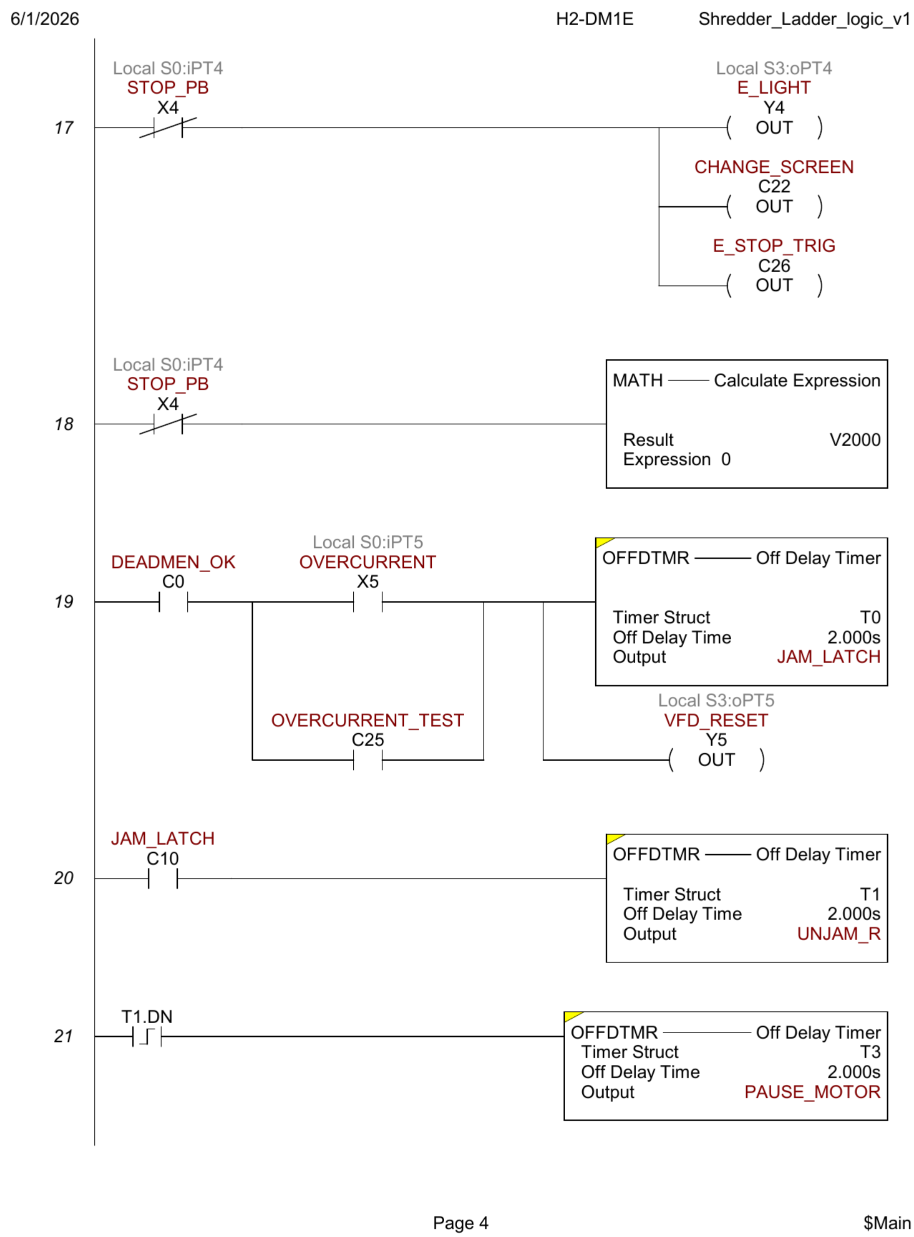

Figure 10: Ladder export page 4. E-stop response, jam detection, and unjam stroke (rungs 17–21).

- **Rung 17** (NOT X4 → OUT Y4 + OUT C22 + OUT C26): when E-stop opens (X4 LOW), immediately light the fault LED, set the screen-change flag for HMI Screen 3, and set the E_STOP_TRIG indicator.
- **Rung 18** (NOT X4 → MATH V2000=0): when E-stop opens, zero the speed setpoint to prevent a speed step on power restore.
- **Rung 19** (C0 AND (X5 OR C25) → OFFDTMR T0 (2 s) → C10 + OUT Y5): jam detection. While deadman is held and the VFD fault relay is active (or HMI soft test is active), JAM_LATCH (C10) goes HIGH immediately and stays HIGH 2 s after the input drops; Y5 (VFD_RESET) is energized directly to pulse the VFD RST terminal.
- **Rung 20** (C10 → OFFDTMR T1 (2 s) → C13): unjam reverse stroke timer. UNJAM_R (C13) stays HIGH for 2 s after JAM_LATCH drops, so rung 10 forces Y7 (reverse) for at least 2 s of recovery.
- **Rung 21** (T1.DN rising edge → OFFDTMR T3 (2 s) → C17): post-reverse pause. When UNJAM_R finishes, T3 holds PAUSE_MOTOR HIGH for 2 s so the motor can coast to rest before re-accelerating forward.

Page 5 of 7 — Rungs 22–26: Accumulating Lockout Timer and Jam Counter

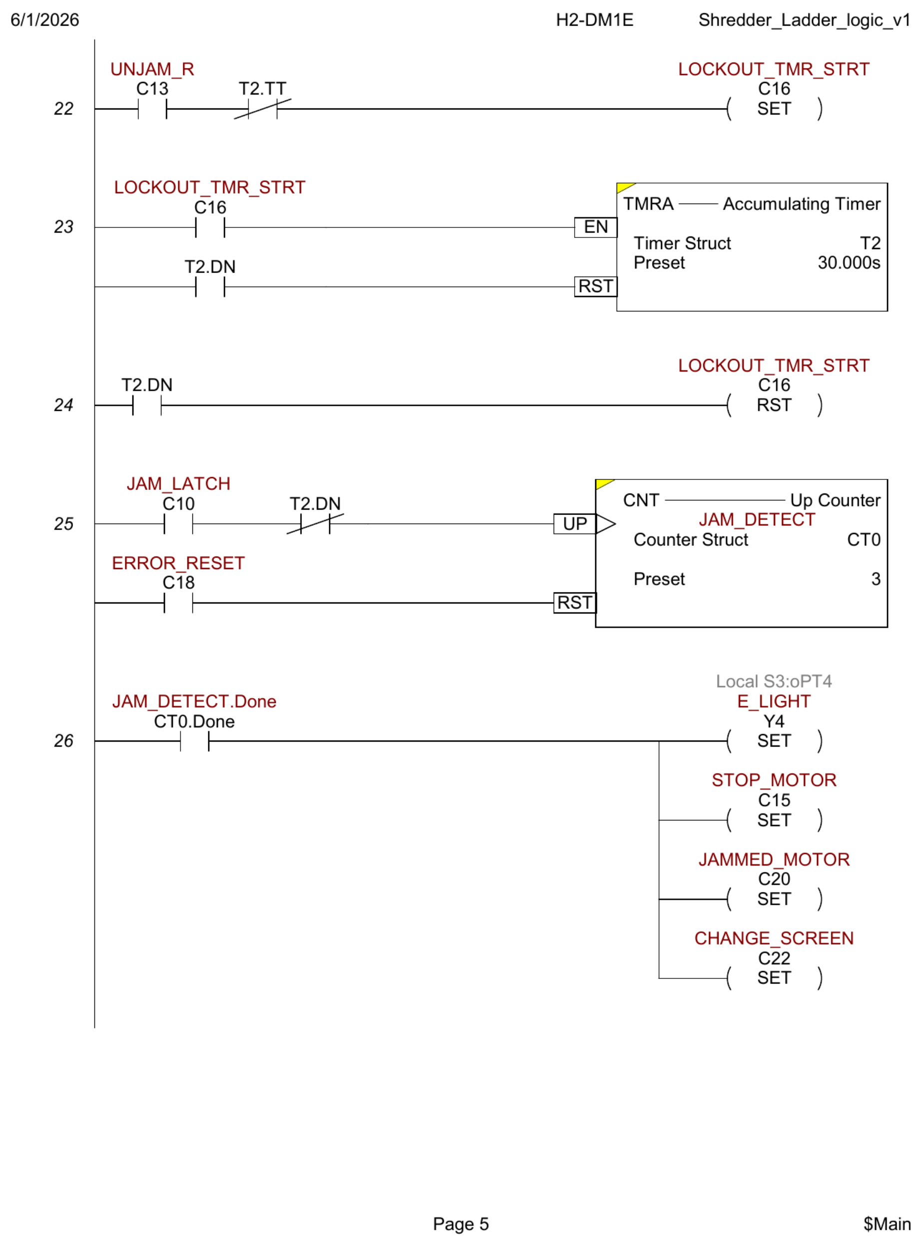

Figure 11: Ladder export page 5. Accumulating lockout timer and jam counter (rungs 22–26).

- **Rung 22** (C13 AND NOT T2.TT → SET C16): starts the jam-count accumulating timer. The NOT T2.TT condition prevents re-triggering if T2 is mid-count.
- **Rung 23** (C16 → TMRA T2 EN; T2.DN → RST T2): 30-second accumulating timer. T2.DN resets itself and rung 24 clears C16, stopping accumulation.
- **Rung 24** (T2.DN → RST C16): after 30 s of accumulated unjam activity, stop accumulation. New JAM_LATCH events no longer increment CT0.
- **Rung 25** (C10 AND NOT T2.DN → CNT UP CT0 (preset 3) [RST: C18]): jam counter increments each time JAM_LATCH goes HIGH while T2 has not completed. Preset 3 means CT0.Done fires after the third jam.
- **Rung 26** (CT0.Done → SET Y4 + SET C15 + SET C20 + SET C22): lockout on third jam. Y4 fault LED on permanently, STOP_MOTOR hard-stops the motor (rungs 9 and 10 block), JAMMED_MOTOR triggers the Screen 3 indicator, CHANGE_SCREEN switches to Screen 3.

Page 6 of 7 — Rungs 27–29: Fault Reset, Capacitive Touch, HMI Screen Change

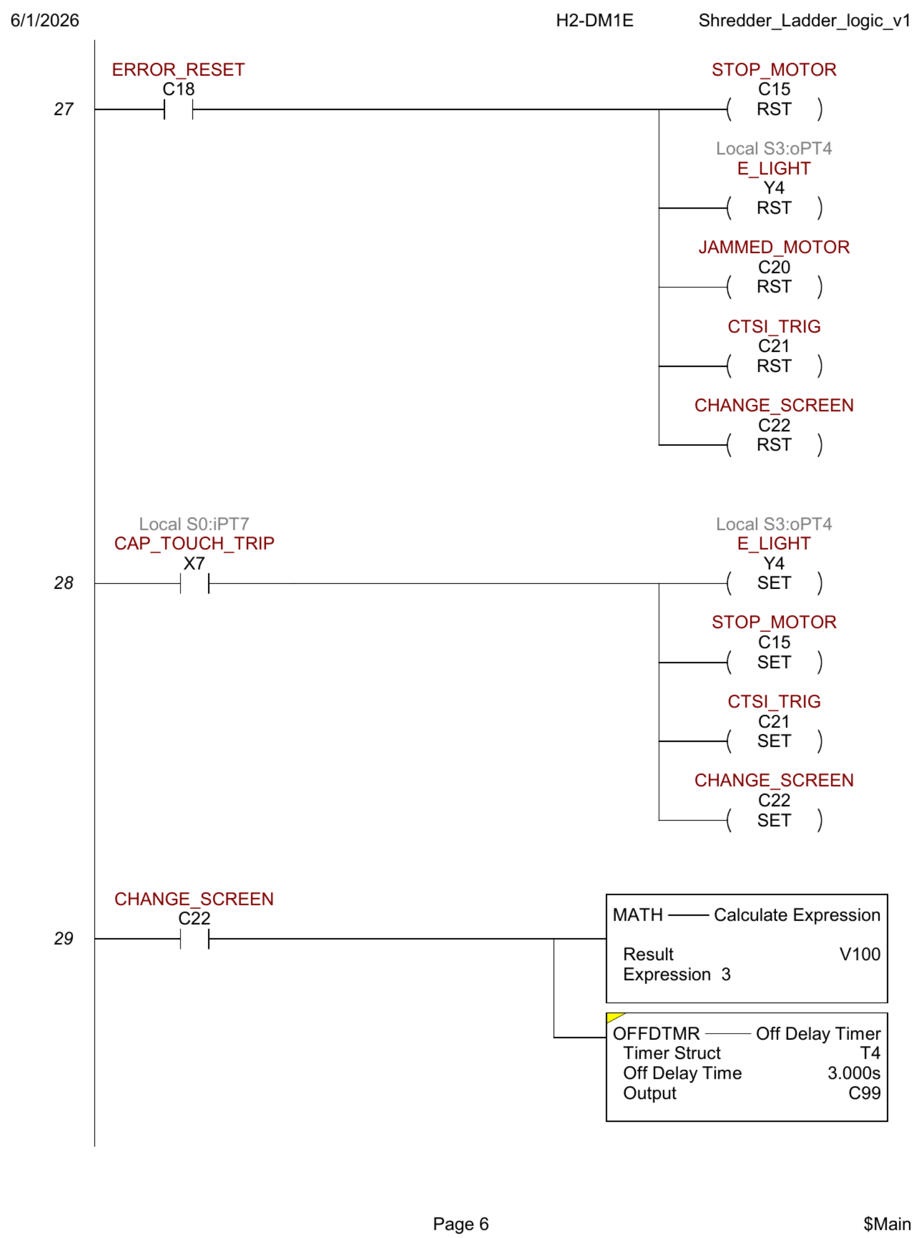

Figure 12: Ladder export page 6. Fault reset path, capacitive touch trip, and HMI screen change (rungs 27–29).

- **Rung 27** (C18 → RST C15 + RST Y4 + RST C20 + RST C21 + RST C22): fault reset path (from HMI Screen 3 SHREDDER JAM or CAP TOUCH button, tag ERROR_OFF = C18).
- **Rung 28** (X7 → SET Y4 + SET C15 + SET C21 + SET C22): capacitive touch safety trip. Any HIGH on X7 immediately sets all four lockout flags.
- **Rung 29** (C22 → MATH V100=3 + OFFDTMR T4 (3 s) → C99): HMI screen navigation. MATH writes 3 to V100 (ERROR_SCREEN tag) so the HMI switches to Screen 3.

Page 7 of 7 — Rungs 30–31: Screen Reset and End of Program

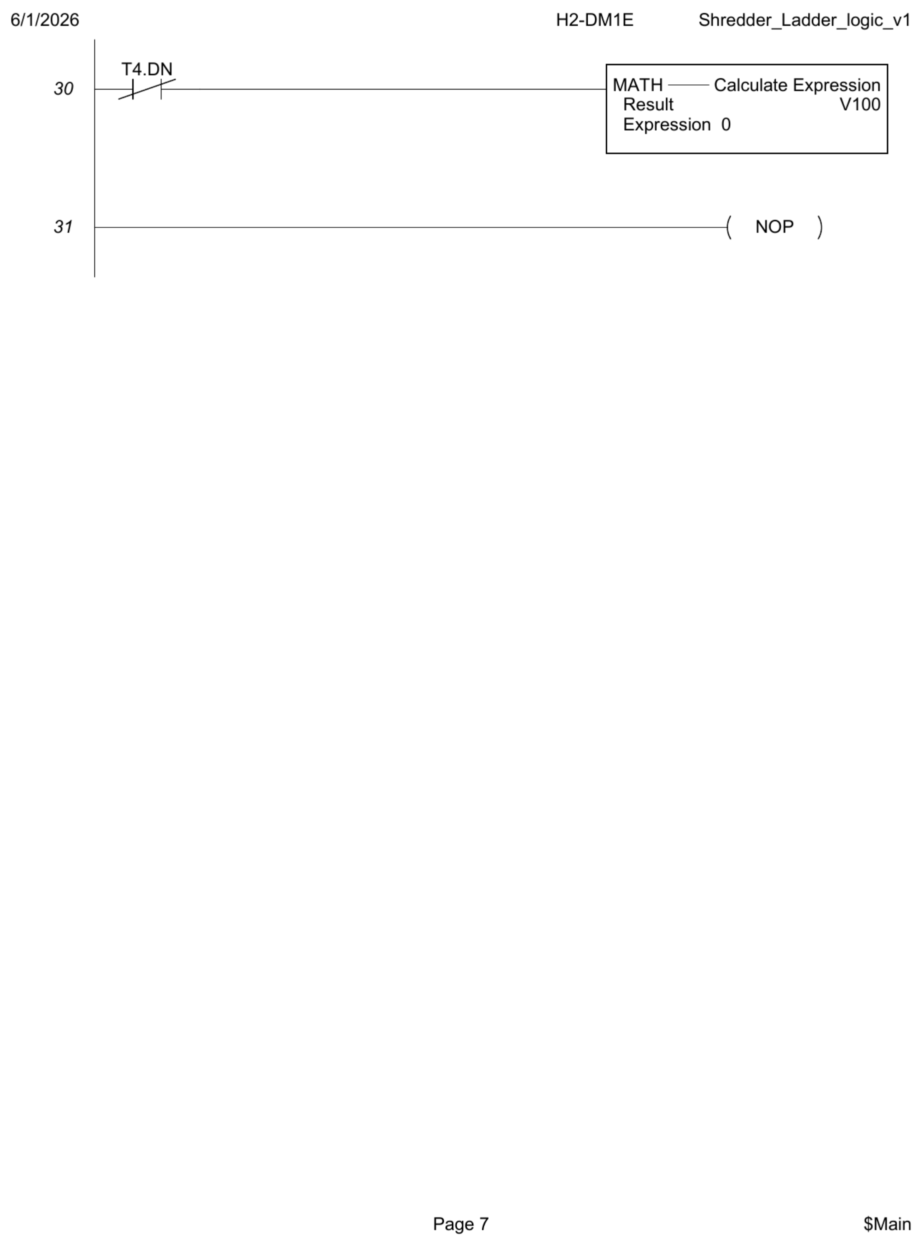

Figure 13: Ladder export page 7. Screen reset timeout and end-of-program NOP (rungs 30–31).

- **Rung 30** (T4.DN → MATH V100=0): after 3 s, V100 is reset to 0, releasing the HMI from the forced screen-3 display.
- **Rung 31** (NOP): no-operation placeholder marking the end of the program.

6.1.5 Jam Recovery Sequence — Complete Timeline

The trace below shows exactly what happens from overcurrent detection through resume, based on the actual timer values and rung logic:

Time	Event / Signal	Active outputs and coils
$t = 0$	X5 (VFD fault relay) goes HIGH while C0 (DEADMEN_OK) is HIGH	T0 input HIGH → C10 (JAM_LATCH) HIGH; Y5 (VFD_RESET) HIGH; C13 (UNJAM_R) HIGH; Y7 (reverse) HIGH; Y6 LOW
$t = 0^+$	VFD fault cleared by Y5 pulse; X5 drops LOW	T0 off-delay starts (2 s); Y5 goes LOW; C10 stays HIGH; C13 stays HIGH; Y7 stays HIGH
$t = 2$ s	T0 off-delay expires → C10 (JAM_LATCH) goes LOW	C10 LOW → T1 off-delay starts (2 s); C13 (UNJAM_R) stays HIGH; Y7 stays HIGH; CT0 increments
$t = 4$ s	T1 off-delay expires → C13 (UNJAM_R) goes LOW; T1.DN rises	Y7 goes LOW (reverse stops); T1.DN rising edge triggers T3; C17 (PAUSE_MOTOR) HIGH
$t = 6$ s	T3 off-delay expires → C17 (PAUSE_MOTOR) goes LOW	Rung 9 NOT C17 condition clears; Y6 (forward) reenergizes if deadman still held
Lockout at $t = 0$ of 3rd jam	CT0.ACC = 3 → CT0.Done HIGH	C15 (STOP_MOTOR) SET; C20 (JAMMED_MOTOR) SET; Y4 SET; C22 → Screen 3. Motor locked until C18 (ERROR_RESET) from HMI.

6.2 HMI Programming

The HMI is an AutomationDirect EA1-T4CL 4-inch C-more Micro-Graphic touch panel running the `HMI_413.mgp` project, connected to the H2-DM1E CPU over the Do-more serial protocol. The project has 4 screens and 27 PLC tags.

6.2.1 HMI Tag Database

Every PLC address visible on the HMI is documented below. Tags without a PLC address are local HMI variables (used for navigation, transient state, or labels).

#	Tag Name	Data Type	PLC Address	What it represents
1	AUTO	Discrete	—	HMI-local; not used in current build
2	CTSI_TRIG	Discrete	C21	Capacitive touch safety trip flag (SET by X7 rung)
3	CURRENT	Signed int 16	V3000	Scaled motor current (0–140 range, from WX0 via rung 3 SCALE)
4	D-MAN A	Discrete	C23	TESTMAN_A_OK — HMI soft deadman A (Test screen)
5	D-MAN B	Discrete	C24	TESTMAN_B_OK — HMI soft deadman B (Test screen)
6	DEADMEN_OK	Discrete	C0	Both deadman buttons (physical or test) held simultaneously
7	ERROR_LIGHT	Discrete	Y4	Error indicator output — physical panel fault LED
8	ERROR_OFF	Discrete	C18	Error reset coil — clears all latched faults when momentarily SET
9	ERROR_SCREEN	BCD int 16	V100	HMI screen selector — written to 3 on fault, 0 after 3 s (T4)
10	E_STOP_TRIG	Discrete	C26	E-stop triggered flag — SET when X4 (STOP_PB) opens
11	FREQ	Signed int 16	V11	VFD output frequency in 0.1 Hz units (read from VFD via serial or calc)
12	FWD	Discrete	C8	FORWARDS_SET — forward direction selected (latching bit)
13	GRAPHIC IND 1	Discrete	—	HMI-local graphic indicator; unused
14	JAMMED_MOTOR	Discrete	C20	Motor jammed and locked out (SET by CT0.Done rung)
15	JAM_COUNTER	BCD int 16	N0	CT0 accumulator — jam count 0–3 (N0 = CT0.ACC in Do-more)

16	MOTOR_STATUS	Discrete	C12	RUN_REQUEST — motor run command active
17	OVERCURRENT	Discrete	C25	OVERCURRENT_TEST — soft overcurrent trigger (Test screen button)
18	RPM	Signed int 16	V14	Calculated or measured shaft RPM (destination register)
19	RPM_CTRL	Signed int 16	V2000	Speed setpoint (0–1600 counts) — scaled to WY0 (0–4095) for VFD analog out
20	RVS	Discrete	C9	REVERSE_SET — reverse direction selected (latching bit)
21	SPEED	Signed int 16	—	HMI-local; unused
22	SPEED(1)	Signed int 16	—	HMI-local; unused
23	STOP	Discrete	—	HMI-local; unused
24	STOP_REQ	Discrete	C1	HMI_STP_REQ — operator stop request from HMI STOP button
25	TEMP	Signed int 16	V15	Temperature register (destination, RTD module — not yet calibrated)
26	VALUE	Signed int 32	—	HMI-local; unused
27	VOLTAGE	Signed int 16	V13	Voltage register (destination — source TBD)

6.2.2 Screen 1 — MOTOR CONTROL

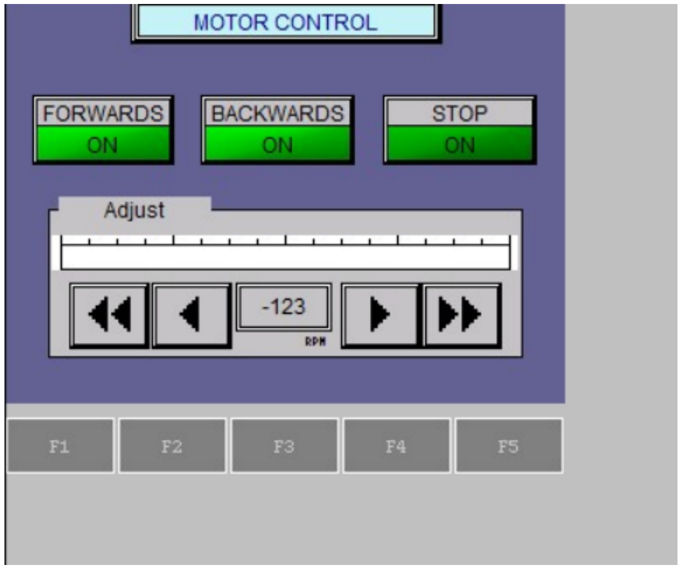

Figure 14: HMI Screen 1 — Motor Control (operator screen).

The primary operator screen. Header bar reads **MOTOR CONTROL**. Contains three toggle pushbuttons and a speed adjust panel.

- **FORWARDS button** (`FWD / C8`, toggle): toggles FORWARDS_SET. ON = green, OFF = dark. Sets C8, resets C9 (rung 13).
- **BACKWARDS button** (`RVS / C9`, toggle): toggles REVERSE_SET. Sets C9, resets C8 (rung 14).
- **STOP button** (`STOP_REQ / C1`, toggle): toggles HMI_STP_REQ. Rung 7 uses NOT C1 to block RUN_REQUEST.
- **Bar graph** (`RPM_CTRL / V2000`): horizontal bar 0–1600 showing current V2000 value.
- **Numeric display** (`RPM_CTRL / V2000`): 3-digit integer labeled 'RPM' below.
- **IncDec ±50 / ±250** (`RPM_CTRL / V2000`): fine and coarse speed adjust buttons.
- **F1/F2/F3 keys**: navigation. F1 → Screen 1, F2 → Screen 2, F3 → Screen 3.

Speed control data path: V2000 (RPM_CTRL, 0–1600) is scaled by rung 2 to WY0 (0–4095), which drives the analog output module F2-08DA-2 CH1 → VFD VI terminal (0–10 V frequency reference).

6.2.3 Screen 2 — MOTOR TELEMETRY

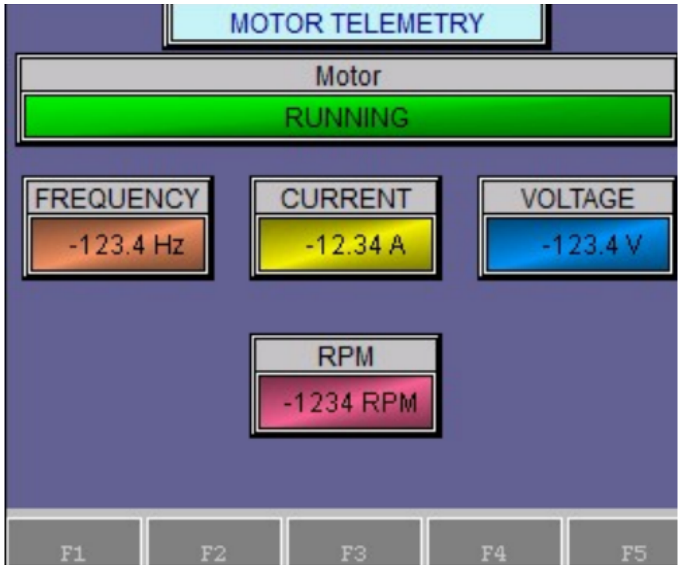

Figure 15: HMI Screen 2 — Motor Telemetry (read-only status).

Header bar reads **MOTOR TELEMETRY**. Read-only status display. A full-width indicator light spans the top and shows RUNNING (green) or STOPPED (gray).

- **Motor status indicator** (`MOTOR_STATUS / C12`): full-width bar, ON = 'RUNNING' green, OFF = 'STOPPED' gray.
- **FREQUENCY display** (`FREQ / V11`): 4-digit, 1 decimal, Hz. Orange background.
- **CURRENT display** (`CURRENT / V3000`): 4-digit, 2 decimal, A. Yellow background.
- **VOLTAGE display** (`VOLTAGE / V13`): 4-digit, 1 decimal, V. Blue background.
- **RPM display** (`RPM / V14`): 4-digit, 0 decimal, RPM. Pink background.
- **F1/F2/F3 keys**: navigation. F3/F4/F10 → Screen 3.

6.2.4 Screen 3 — ERROR (Fault / Alarm Screen)

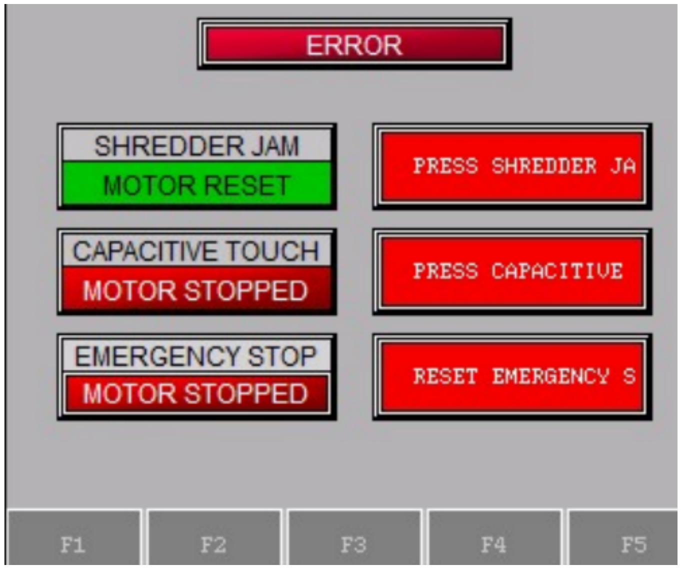

Figure 16: HMI Screen 3 — Error / Alarm screen (red backlight, three fault rows).

Backlight color 11 (red backlight). Automatically shown when CHANGE_SCREEN (C22) is SET by the ladder (rung 29 writes V100=3; after 3 s T4 resets V100 to 0). Three fault rows each with a status indicator and scrolling instruction text.

- **ERROR header indicator** (`ERROR_LIGHT / Y4`): ON shows 'ERROR' in red; OFF shows 'ERROR RESET' in gray. Y4 is the physical fault LED output.
- **SHREDDER JAM row** (button + indicator): button `ERROR_OFF / C18`; indicator `JAMMED_MOTOR / C20`. Indicator lit red ('MOTOR STOPPED') when C20 HIGH (CT0.Done fired). Button pulses C18 (ERROR_RESET).
- **CAPACITIVE TOUCH row** (button + indicator): button `ERROR_OFF / C18`; indicator `CTSI_TRIG / C21`. Indicator lit red when C21 HIGH (capacitive touch trip).
- **EMERGENCY STOP row** (indicator only): `E_STOP_TRIG / C26`. ON shows 'MOTOR STOPPED' red; OFF shows 'MOTOR RESET' green.
- **Scroll text**: each row scrolls instructions: 'PRESS SHREDDER JAM BUTTON TO RESET MOTOR.', 'PRESS CAPACITIVE TOUCH TRIGGER BUTTON TO RESET MOTOR.', 'RESET EMERGENCY STOP BUTTON TO RESET MOTOR.'

6.2.5 Screen 4 — TEST (Engineering / Commissioning)

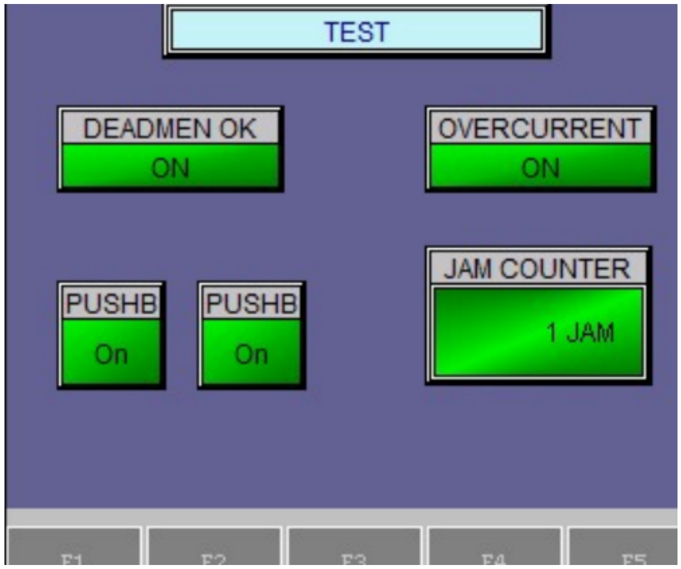

Figure 17: HMI Screen 4 — Test / Commissioning (soft inputs for bench testing).

Header reads **TEST**. Used during commissioning to simulate inputs without the physical pushbuttons. Not accessible from normal operation screens (only reachable via F5 from Screen 3).

- **DEADMEN OK button** (`DEADMEN_OK / C0`, toggle): directly forces C0 HIGH. Allows testing run logic without holding physical buttons.
- **OVERCURRENT button** (`OVERCURRENT / C25`, momentary on): pulses C25 (OVERCURRENT_TEST). Rung 19 treats C25 same as X5, triggering the full jam recovery sequence.
- **JAM COUNTER display** (`JAM_COUNTER / N0`, BCD): shows CT0 accumulator as jam count 0–3.
- **PUSHBUTTON A** (`D-MAN A / C23`, toggle): forces C23 (TESTMAN_A_OK) HIGH. Rung 5 uses C23 as alternate to X2 for deadman A.
- **PUSHBUTTON B** (`D-MAN B / C24`, toggle): forces C24 (TESTMAN_B_OK) HIGH. Rung 5 uses C24 as alternate to X3 for deadman B.

7 Safety Design

7.1 NFPA 79 Compliance

- Wire color coding follows NFPA 79 §13.2: Black for AC line/load, Red for AC control, Blue for DC control, Blue/W stripe for DC return, Green/Yellow for ground.
- This color convention is applied to every conductor in the panel.
- All control wires labeled at both ends.
- Control wires land as straight stripped conductors at the terminal blocks (no ferrules used in this build).

7.2 Safety Functions

- **E-stop:** Category 0 stop (immediate removal of power). Hardwired — an E22B1 NC contact in series with the SD-N35 contactor coil; pressing it drops the coil and opens the contactor independently of the PLC. Single channel; the PLC additionally monitors the state on X4 (not a safety-rated channel).
- **Two-hand / deadman control:** both blue pushbuttons must be held continuously to enable the motor. As-built, the two NO inputs are AND-ed in the standard PLC's ladder logic to gate the VFD run output. It is the run-permissive and the safety interlock satisfying requirement M6, but it is *not* a safety-rated two-hand device per ISO 13851 Type III (no simultaneity timer, complementary contacts, or self-monitoring).
- **Access guarding:** handled by the ME team's mechanical design (feed geometry), so no electrical lid interlock is implemented (a lid interlock was a may-have, not a requirement).
- **Capacitive-touch module (planned):** a capacitive-touch safety module is on hand and the ladder logic to act on it (input X7 → stop motor + fault indication) is written, but it is *not yet wired*. This is a may-have (additional safety sensing) and is not relied upon in the current build.
- **Overload:** the VFD's electronic motor-overload function plus the over-torque trip (see Jam Detection); equipment protection rather than a personnel-safety function.

7.3 Performance Level (ISO 13849-1) — Preliminary Assessment

This is a preliminary, self-assessed determination from the as-built architecture. A formal verification needs a documented ISO 12100 risk assessment (to set the required performance level, PLr) and component reliability data (MTTFd / B10d, diagnostic coverage). Table 2 summarizes the achieved categories.

Table 2: Preliminary ISO 13849-1 assessment of the as-built safety functions.

Safety function	As-built architecture	Category (prelim.)
Emergency stop	Hardwired E22B1 NC contact in series with the SD-N35 contactor coil; opening it removes motor power. Single channel, well-tried components; PLC monitors X4 (not safety-rated).	Cat 1 (PL b–c)
Two-hand / deadman	Two NO buttons AND-ed in the standard PLC to gate the VFD run output. Single channel; no simultaneity check, complementary contacts, or self-monitoring.	Cat B (PL b); not ISO 13851 Type III
Over-torque / jam stop	VFD over-torque trip plus electronic motor overload. Equipment protection, not a personnel-safety function.	Not rated (process protection)

Overall the control system is approximately **Category B–1 (PL b–c)**, limited by: single-channel architecture throughout; a standard (non-safety) PLC evaluating the two-hand permissive; and a single contactor with no force-guided (mechanically linked) contacts and no feedback (EDM) monitoring to detect a welded contactor. The hardwired E-stop, built from well-tried components, is the strongest function (Category 1).

To raise the achieved PL — for example to Category 3 / PL d, should the risk assessment require it for the rotating-blade hazard — would need a dual-channel architecture, a safety relay or safety-rated PLC, two contactors (or force-guided contacts) with feedback monitoring, and a synchronous two-hand module meeting ISO 13851.

7.4 Failure Mode Analysis

- Wire break in jam signal: PLC sees X5 stuck LOW. Jam signal cannot fire, but the hardwired E-stop loop and motor contactor provide independent safety. Operator can also see motor stop and trigger investigation via fault-light absence.
- VFD loses power: the FA–FC fault relay de-energizes and opens, so X5 reads LOW (indistinguishable from healthy). The motor stops because the VFD itself is unpowered, so the condition is caught by the motor not running rather than by the jam signal.
- PLC fails: motor contactor drops out (safety chain bypasses PLC). Motor coasts to stop.
- Brake resistor disconnected: VFD will trip on overvoltage during deceleration. Motor coasts to stop. Diagnosable from VFD display.

8 User Manual

8.1 Purpose of This Manual

This manual provides operators, maintenance personnel, and safety officers with the procedures necessary to safely operate, monitor, and perform basic troubleshooting on the Universal Plastic Shredder V2.0 electrical control system. It assumes the user has read the full engineering report and understands the basic hazards described in Section 7 (Safety Design).

8.2 Intended Audience and Required Training

- **Operators:** Must complete a site-specific safety briefing covering two-hand deadman operation, E-stop use, and jam recovery. No electrical troubleshooting is permitted without additional training.
- **Maintenance personnel:** Must be familiar with NFPA 70E arc-flash boundaries, lockout/tagout (LOTO) per 29 CFR 1910.147, and the contents of Appendix B (I/O point list) and Appendix D (VFD parameters).
- **Supervisors / safety officers:** Shall maintain a copy of this manual and the full engineering report adjacent to the machine or in a controlled electronic repository.

8.3 Normal Operating Procedure

8.3.1 Pre-Start Checks

1. Verify the main disconnect (PowerTec 71008) is in the OFF position.
2. Inspect the feed area for obstructions, loose plastic, or foreign objects.
3. Verify the E-stop button is pulled out (released).
4. Ensure both deadman pushbuttons are not pressed.
5. Close and latch any mechanical access guards (ME team's responsibility).

8.3.2 Startup

1. Turn the main disconnect to ON. The HMI shall display **IDLE / READY**. The green status indicator shall illuminate.
2. If the HMI shows a fault or lockout message, press the HMI **RESET** button (see Section 8.6).
3. To start shredding:

- Press and **hold both blue deadman pushbuttons simultaneously**.
- The motor will accelerate to the programmed speed (70–90 RPM at the shaft).
- The red "RUNNING" status indicator shall illuminate.
- If either deadman button is released, the motor stops immediately.

4. Feed plastic sheet material at a moderate, steady rate. Avoid overstuffing.

8.3.3 Normal Shutdown

1. Release both deadman buttons. The motor will coast to a stop.
2. Wait for the blades to come to a complete stop (minimum 5 seconds).
3. Turn the main disconnect to OFF.

8.4 Emergency Stop and Safety Functions

- **To stop in an emergency:** Press the red mushroom E-stop button. Motor power is removed immediately (Category 0 stop). The HMI will show **ESTOP ACTIVE**.
- **To reset after E-stop:** Twist the E-stop button clockwise to release it, then press the HMI RESET button.
- **Two-hand deadman:** The motor runs only while both blue buttons are held. This is a permissive — not a safety-rated two-hand control per ISO 13851 Type III. Never bypass or defeat this function.

8.5 Jam Detection and Automatic Recovery

The control system includes an auto-reverse jam recovery feature. When an over-torque jam is detected (motor current exceeds 150% of rated for 3 seconds):

1. The VFD trips and the motor stops.
2. The HMI displays **JAM DETECTED** and increments the jam counter.
3. The PLC automatically:
   (a) Sends a 500 ms reset pulse to clear the VFD fault.
   (b) Runs the motor in reverse for 2 seconds to clear the jam.
   (c) Pauses briefly, then restarts forward operation.
4. If the jam clears, normal operation resumes.

5. If a jam occurs again within the same operating session:
   - Jam counter increments.
   - After the **third jam** within a session, the system enters **LOCKOUT** mode.
   - The HMI shows **LOCKOUT — MANUAL RESET REQUIRED**.
   - Motor cannot restart until a supervisor presses the HMI RESET button and clears the feed area.

8.6 Fault Recovery and Lockout Reset

Table 3: Fault codes and recovery actions

HMI Display	Operator Action
ESTOP ACTIVE	Release E-stop button, then press HMI RESET.
VFD FAULT	Check VFD display for code (overvoltage, overtemp, etc.). Power cycle main disconnect. If repeated, call maintenance.
JAM (1 or 2)	No action — automatic recovery runs. Monitor HMI.
LOCKOUT (3 jams)	Clear plastic from blades manually (after LOTO), then press HMI RESET.
IDLE / READY	Normal.

8.7 Schematic and Wiring Diagrams

The following figures show the core electrical circuits of the shredder control system. All wire colors follow NFPA 79 §13.2 (see Section 7.1). Full terminal-level details are in Appendix B (I/O point list).

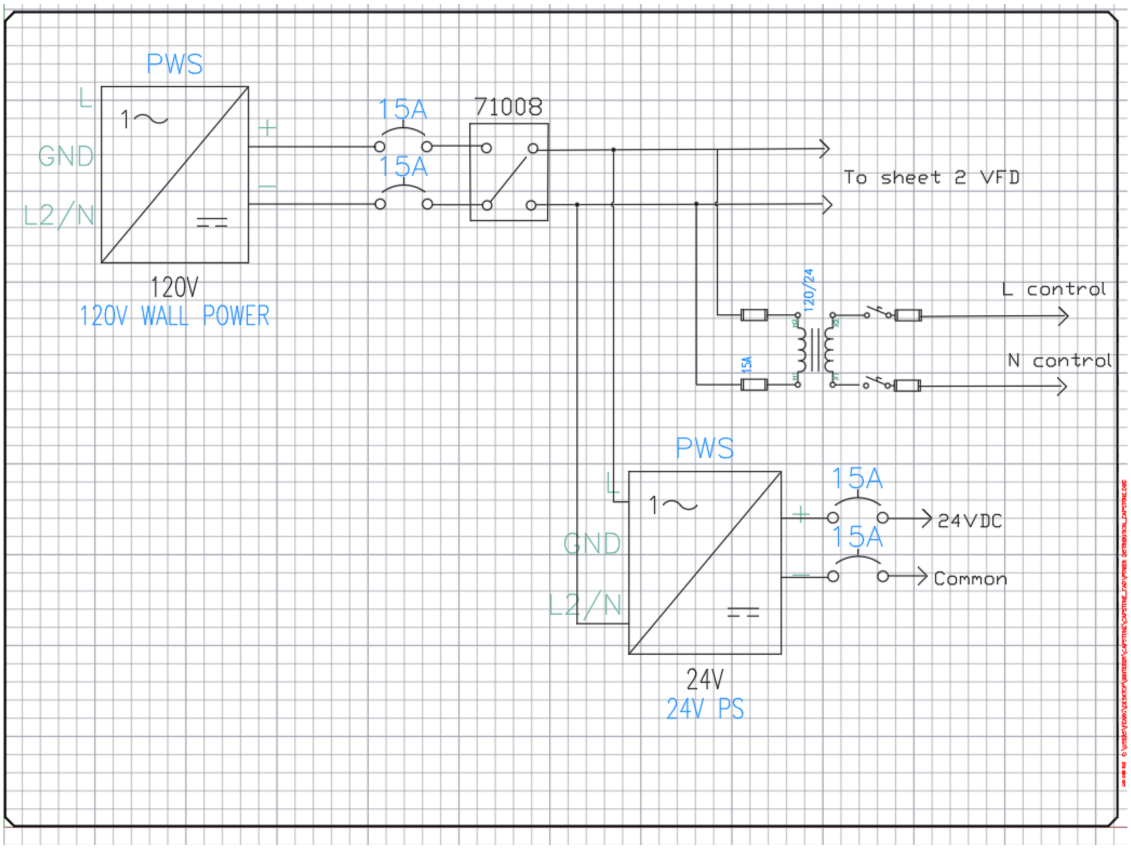

Figure 18: Power distribution circuit: main disconnect, branch breakers, 24 V power supply, and VFD input.

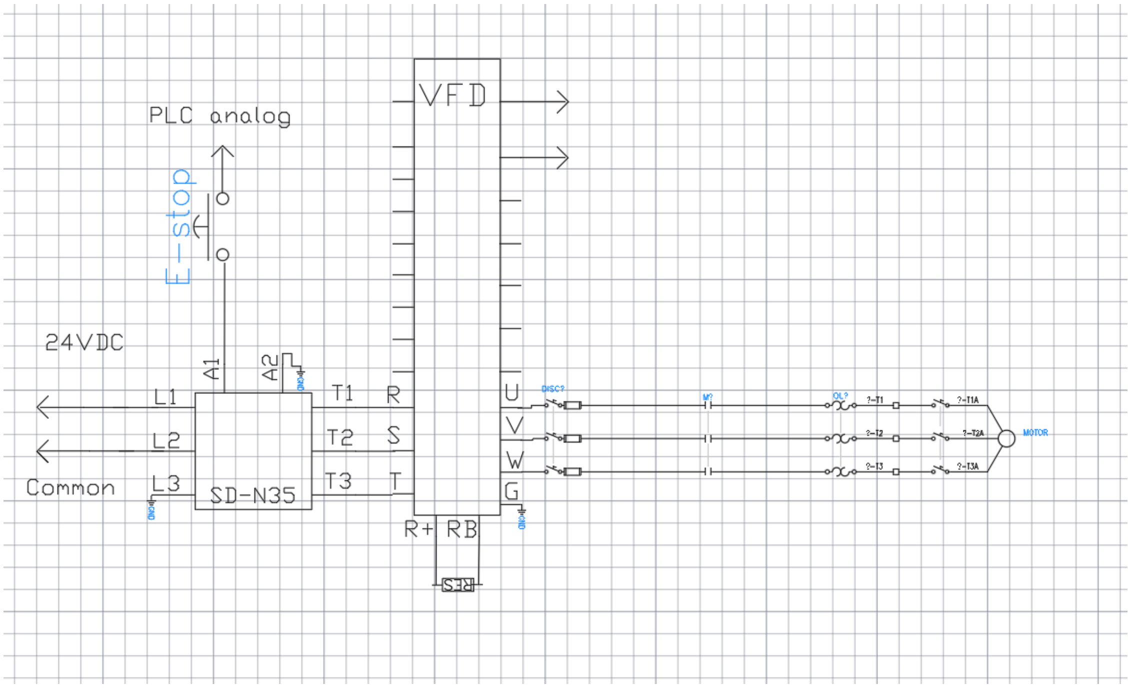

Figure 19: Motor power circuit: 24 VDC E-stop chain feeding the SD-N35 contactor coil; AC service (L1 / L2 / Common) through the contactor pole into the VFD (R / S / T); VFD outputs U / V / W to the motor through the panel-mount disconnect and the overload; brake resistor on R+ / RB.

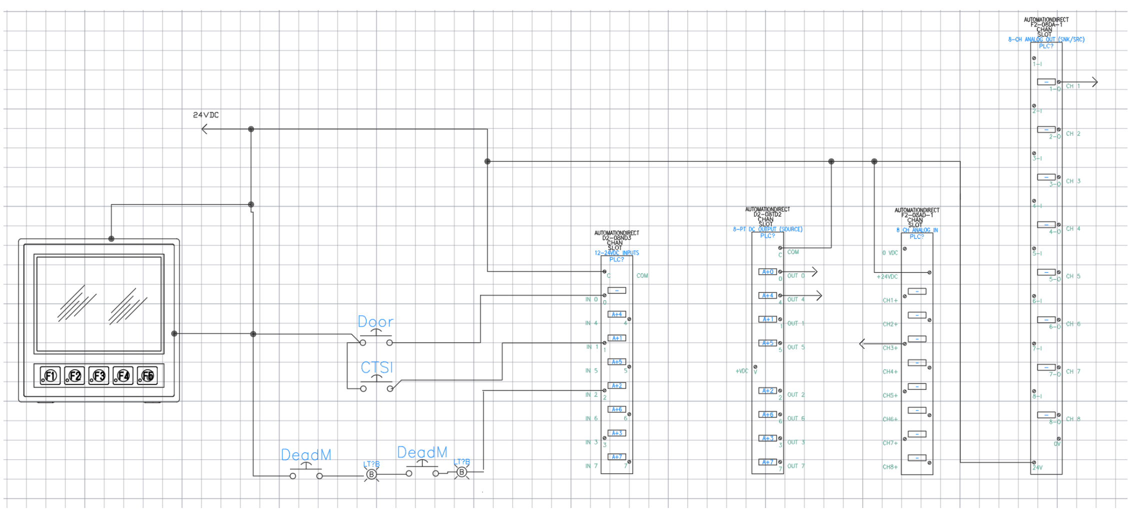

Figure 20: PLC I/O circuit: 24 VDC distribution to the HMI, the two deadman pushbuttons, the door / interlock taps, and the PLC discrete-input terminal blocks. Includes the E-stop monitor tap, the VFD fault input, and the VFD run / reverse / reset relay outputs.

8.8 Daily and Periodic Maintenance

8.8.1 Operator Daily Checks

- Before first use: Verify E-stop button stops the motor immediately.
- Check that both deadman buttons are responsive (motor runs only when both held).
- Listen for unusual noises (grinding, excessive vibration).
- Ensure the HMI jam counter resets to zero after a clean 30-second run.

8.8.2 Monthly Maintenance (Qualified Personnel Only)

- Perform LOTO (lockout/tagout) on the main disconnect.
- Open the electrical enclosure.
- Visually inspect for loose wires, discoloration, or burned terminals.
- Check that the brake resistor (75 ohm, 500w) shows no cracking or overheating signs.
- Verify fan operation on the VFD and power supply.
- Clean dust from the enclosure using low-pressure compressed air.
- Close and secure enclosure before re-energizing.

8.9 Troubleshooting Guide

Table 4: Common issues and corrective actions

Symptom	Possible cause	Action
Motor does not start	E-stop pressed	Release E-stop, reset HMI.
	Only one deadman pressed	Press both simultaneously.
	LOCKOUT active	Clear jam, press HMI RESET.
Motor runs briefly then stops	Jam detected	Let auto-reverse run. If repeated, clear plastic manually.
HMI blank	No 24V power	Check Mean Well PSU input breaker (10A) and output fuses.
VFD shows fault code	Overvoltage / overtemp / etc.	See VFD manual Appendix D. Power cycle.
Jam counter increments without jam	Over-torque threshold too low	Check PD124. Nominal = 150%.

8.10 Safety Warnings Summary

CRITICAL SAFETY WARNINGS

- Never reach into the feed area while blades are moving. Wait for complete stop.
- Do not bypass, tape down, or modify the deadman buttons or E-stop.
- Lockout/tagout the main disconnect before any enclosure opening or blade service.
- This machine has no electrical feed interlock — the mechanical feed geometry is the primary access safeguard.
- If the automatic jam recovery cycles three times, **stop and manually clear** the jam.

8.11 Contact for Service and Support

For maintenance beyond this manual, contact the Ichor Systems facilities team or the original design team (via Portland State University, ECE Department). Have the following ready:

- Shredder serial number (TBD — assign upon final assembly)
- VFD fault code (if any)
- HMI jam count and lockout status
- Photos of any visible damage

8.12 Revision History of This Manual

Version	Date	Changes
1.0	2026-05-31	Initial release (rough draft).

9 Standards and Regulatory Compliance

This section consolidates every electrical and machine-safety standard the design is built against and states, for each, how it is applied to the shredder. Many of these are referenced in passing elsewhere in this report; they are gathered here so a reviewer (or a future maintenance team) can audit compliance in one place.

9.1 Standards Summary Matrix

Standard	Scope	How it is applied here
NFPA 70 (NEC)	National Electrical Code – premises wiring, motor circuits	Branch-circuit, conductor, grounding, and disconnect design (see §9.2).
NFPA 79	Electrical standard for industrial machinery	Wire color coding, supply disconnect, protective bonding, stop functions, operator devices (see §9.3).
NEMA 250	Enclosure type ratings	PLC/HMI panel housed in a Vorksmen SEB001 steel control enclosure, manufacturer-rated NEMA Type 12 (dust-tight, drip-tight); meets the lab/shop environment requirement (see §9.4).
NEMA ICS 1/2	Industrial control & systems	Contactor / control device selection and marking.

continued on next page . . .

Standard	Scope	How it is applied here
NEMA MG 1	Motors and generators	Motor nameplate interpretation and VFD parameterization.
UL 508A	Industrial control panels	Panel construction practice and short-circuit current rating (SCCR) awareness.
IEC 60204-1	Safety of machinery – electrical equipment	International counterpart to NFPA 79; used to cross-check stop categories.
ISO 12100	Risk assessment / risk reduction	Framework behind the hazard analysis driving the safety circuit.
ISO 13849-1	Safety-related parts of control systems (PL)	Preliminary self-assessment (Cat B–1) of the E-stop and two-hand control chain.
ISO 13850	Emergency stop function	E-stop actuator type, color, and Category 0 behavior.
ISO 13851	Two-hand control devices	Deadman two-hand start timing and configuration.
OSHA 29 CFR 1910.147	Lockout/Tagout (LOTO)	Lockable main disconnect for energy isolation during service.

9.2 NFPA 70 – National Electrical Code

NEC article	Requirement	How it is applied here
Art. 430.6	Motor FLC basis	From the GE test-motor nameplate (FLA 7.6 A at 230 V); basis for all downstream sizing. Re-verified when the production motor is selected.
Art. 430.22	Branch conductor sizing	Conductors sized at ≥ 125% of motor FLC.
Art. 430.32	Running overload	VFD electronic motor-overload (`PD142 = 7.6 A`) plus the over-torque trip.
Art. 430.52	Branch short-circuit / ground-fault	30 A C-curve breaker on the VFD branch; clears inrush without nuisance trips (inverse-time-breaker allowance).
Art. 430.102/.109	Disconnecting means	Provided ahead of the controller and VFD.
Art. 430 Part X (430.122)	Adjustable-speed drive systems	Conductor and protection rules applied to the VFD-fed motor circuit.
Art. 250	Grounding & bonding	EGC (green / green-yellow) bonds the VFD, PSU, motor frame, and panel to a common ground bus and building earth.
Art. 310	Conductor ampacity	Conductors chosen from the ampacity tables for the installed temperature rating.
Art. 409	Industrial control panels	Panel built to Art. 409 marking and overcurrent rules: every conductor labeled at both ends, every load individually fused at its terminal block, and the disconnect-ahead-of-the-controller arrangement is field-marked at the enclosure door.
Art. 110.26	Working space	Clearances around live parts maintained for safe service; final panel mounting confirmed to leave the required clearance.

9.3 NFPA 79 – Electrical Standard for Industrial Machinery

NFPA 79 clause	Requirement	How it is applied here
§5.3	Supply disconnecting means	Lockable main disconnect (PowerTec 71008) isolates the machine from supply; also satisfies LOTO.
§7	Overcurrent protection	Branch and downstream fusing at terminal blocks (PSU 25 A slow-blow, PLC 1 A fast-blow, control loads individually fused).
§8	Protective bonding circuit	Continuous equipment-grounding / bonding path to all exposed conductive parts.
§9	Control circuits & functions	Stop functions classified by stop category; the hardwired safety chain has precedence over the PLC.
§10.7–10.8	Operator interface / E-stop	Red mushroom actuator on a yellow background, self-latching NC contact; start devices guarded by the two-hand requirement.
§13.2	Conductor color coding	Black AC line/load, red AC control, blue DC control, blue/white DC return, green/yellow PE; wires labeled both ends, landed straight (no ferrules in this build).

9.4 NEMA Standards

Standard	Covers	How it is applied here
NEMA 250	Enclosure type ratings	PLC / HMI control box is a Vorksmen SEB001 steel enclosure, manufacturer-rated NEMA Type 12 (dust-tight, drip-tight), sealed and gasketed to keep airborne particulate out of the panel in the lab/shop environment. Sourcing entry is in Appendix A (BOM); see also Section 12 (References).
NEMA ICS 1 / ICS 2	Industrial control & systems	Mitsubishi SD-N35 contactor selected per ICS 1 / ICS 2: AC-3 (squirrel-cage motor) rating 35 A is well above the test motor's 7.6 A FLA, the coil voltage matches the 24 VDC panel control voltage, and the device is permanently labeled at its terminal block.
NEMA MG 1	Motors & generators	Motor nameplate (GE 5KE182BC205B: 230 V, 7.6 A FLA, 4 poles, 1750 RPM) transcribed directly into VFD parameters `PD141 = 230 V`, `PD142 = 7.6 A`, `PD143 = 4`, `PD144 = 1750 RPM` so the drive's slip estimator and electronic overload reference the actual machine.

9.5 Machine-Safety Standards (ISO / IEC)

Standard	Covers	How it is applied here
ISO 12100	Risk assessment / reduction	Identifies the primary hazards (rotating blades, entanglement, electrical) and drives the layered safeguards.
ISO 13849-1	Safety-related control parts (PL)	Preliminary assessment: hardwired E-stop ≈ Cat 1; two-hand permissive ≈ Cat B (see the Safety Functions section).
ISO 13850	Emergency stop function	Category 0 stop (immediate removal of power) with fail-safe NC wiring.
ISO 13851	Two-hand control devices	Both buttons actuated within 0.5 s of each other and held continuously to enable the motor.
IEC 60204-1	Electrical equipment of machines	International analogue of NFPA 79; used to cross-check stop categories and supply disconnection.

9.6 Occupational Safety (OSHA)

Regulation	Covers	How it is applied here
29 CFR 1910.147	Lockout/Tagout (LOTO)	The main supply disconnect is lockable for positive isolation during blade-area service and maintenance.
29 CFR 1910 Subpart S	Electrical safety	General electrical-safety practices (guarding of live parts, grounding) observed throughout the panel build.

9.7 Panel Enclosure

The PLC / HMI control panel is housed in a **Vorksmen SEB001** steel industrial enclosure (sourcing entry in Appendix A, BOM). The enclosure houses the entire electrical system: the DirectLOGIC 205 rack with the H2-DM1E CPU and its five I/O modules, the EA1-T4CL HMI, the Mean Well NDR-480-24 24 VDC supply, the Mitsubishi SD-N35 contactor, the Huanyang HY02D211B-T VFD, the operator pushbutton bank, and the labeled terminal blocks.

Manufacturer specification:

- NEMA Type rating: **Type 12** (dust-tight, drip-tight, indoor industrial use).
- Material: steel (welded seams, gasketed lid).
- Door: hinged, latching, lockable.
- Mounting: floor / wall mount with rear flange.

Verified at installation: the lid gasket is intact, the door latches engage, and Article 110.26 working clearances around the panel are met at the as-built mounting location. The manufacturer datasheet is referenced in Section 12 (References, Manufacturer Datasheets).

10 Testing and Validation

10.1 Signal Chain Tests

- Manual jumper test (FA-FC short): confirmed PLC X5 reads LOW correctly – wiring intact.
- Guaranteed-trip test (`PD124 = 1`, `PD125 = 5.0`): VFD trips reliably on motor run, fault relay fires, PLC X5 drops.
- Threshold tuning: 150% (`PD124 = 150`) chosen for production based on no-load current (0.5 A) versus expected jam current (above 11 A actual).

10.2 Load Testing

Blocked — ME team dependency. Loaded shred testing against representative PVC, PP, and LDPE sheet stock was planned but is not possible from the EE side: the ME team's shredder head is not functional as of the writing of this report, so the machine cannot be assembled, the panel cannot be mounted to the frame, and there is no real cutting load against which to spin the motor. The control system has been validated end-to-end on the bench (signal-chain tests above, plus the parameter / wiring tables in Appendices A–E); when the mechanical build is delivered, the open-work punch list (`docs/term2_week11_Open_Work.md`, item 5) lays out the loaded shred run, the real-jam over-torque re-tune for `PD124` / `PD125`, and the CTSI trip-zone re-validation against the actual cutting geometry.

10.3 Safety Validation

Functional behavior of the hardwired safety chain was confirmed at the bench: pressing the E-stop drops the SD-N35 contactor coil and removes VFD neutral within one contactor open-time; releasing either deadman pushbutton mid-run drops the run-permit and stops the motor; PLC `X4` tracks the contactor-coil node so the ladder observes (but cannot defeat) the safety state. Quantitative characterization of the E-stop response time and the two-hand-release timing with a scope was not completed before end of term and is logged on the open-work punch list as bench-doable follow-up (`docs/term2_week11_Open_Work.md`).

11 Results

11.1 What Works

- All motor parameters programmed and verified.
- Over-torque detection chain validated (VFD detects, relay fires, PLC reads, ladder reacts).
- Fail-safe NC wiring confirmed – system correctly enters fault state on signal chain failure.
- HMI communicates with the PLC over the serial COM link.

11.2 Open Issues

- Vendor-specific Huanyang firmware quirk: `PD052 = 12` does not fire relay in continue-running mode. Working around with `PD052 = 02` + `PD123 = 3`.
- PLC ladder logic not fully complete – state machine implemented, but unjam routine timing needs tuning with real loaded motor.
- Mechanical integration pending – have not yet tested under real shredding load.

- Motor-current monitor (VFD `VO` → F2-08AD-1 CH1) is wired, but under time constraints it was never configured/scaled correctly to read the VFD output current. The signal path exists; the VFD analog-output and PLC analog-input scaling still need to be set and calibrated.

The Huanyang firmware quirk above (`PD052 = 12` continue-running over-torque indication does not fire the relay on our revision) is documented in one line in the Open Issues list and in the note attached to Table 1; the broader design lesson (verify vendor behavior on the bench before relying on the manual) is recorded in Lessons Learned. Kept brief here so it does not crowd out the high-level results.

12 Discussion

12.1 Design Tradeoffs

- Cost versus reliability: chose Huanyang VFD ($150) over ATO ($450). Firmware quirks documented in this report (the Software Design note and Appendix D) for future maintenance.
- 110V single-phase versus 220V split-phase mains: 110V chosen for receptacle availability in PSU lab; trade-off is double the input current and need for a dedicated 30 A circuit.
- NO active-high versus NC fail-safe jam signal: the VFD fault relay is wired NO active-high (FA–FC closes on trip, X5 HIGH). Simpler to wire, but not fail-safe at the signal level—a broken wire reads as "no fault." That residual risk is covered by the independent hardwired E-stop loop and motor contactor, which remove power regardless of the PLC.

12.2 Lessons Learned

- Always verify VFD relay function with multimeter before assuming firmware works as documented.
- Document deviations from datasheet behavior in version control so future teams can find them.
- Power cycle after parameter changes – some VFDs require it for save to fully take effect.
- Use ferrules on control wires in future builds: this build landed straight stranded wire at the terminal blocks, and loose/spread strands caused multiple debugging hours during commissioning.

12.3 Project Management Notes

The team ran on a weekly cadence with three standing touchpoints. Internal working meetings were held once a week to plan tasks, hand off blocking items between members, and review the running TODO list. Faculty review with Dr. Jonathan Bird (EE advisor) ran on either Thursday or Tuesday each week depending on lab and class conflicts; these sessions focused on technical direction (VFD parameterization, jam recovery design, safety architecture) and caught issues like the over-torque relay firmware quirk and the two-hand permissive scope decision before they propagated into the build. Industry coordination with Bridget Lannigan at MME / Ichor Systems happened on Wednesdays and was the primary forum for ME/EE integration, requirements clarifications, and confirming the as-built scope against the sponsor's needs. Prof. Andrew Greenberg attended the broader capstone review checkpoints across both terms.

All source artifacts (this report, the Do-more ladder `.dmd` project and its `Circuit_Logic.pdf` export, the C-more Micro `HMI_413.mgp` project, the AutomationDirect Point List workbook, the BOM workbook, the AutoCAD Electrical project, the wiring documentation, and every supporting markdown reference) live in a single GitHub repository. Tracking the full edit history in version control gave the team a shared audit trail and a working backup, and it is what makes the rung-by-rung walkthrough in Section 6.1 reproducible: every figure in this report points back to a committed source file.

Overall, the schedule held to the rough-draft milestone. VFD selection and parameterization landed in Winter term; panel build, point-list finalization, and the ladder logic finished in early Spring; HMI programming and the safety integration filled the back half of Spring. Items that slipped (final production motor selection, capacitive-touch wiring as a may-have, and inline insertion of the exported AutoCAD wiring diagrams) are tracked as `\todo` markers throughout this document so they stay visible until closed.

13 Future Work and Improvements

13.1 Safety and Compliance

- **Integrate the capacitive-touch safety (CTSI) module:** wire the on-hand capacitive-touch module to input X7. The ladder logic to stop the motor and raise a fault on a touch trip is already written, so this is primarily a wiring and validation task. It adds a no-contact operator-presence safeguard at the feed.

- **Raise the safety performance level:** move from the current single-channel Cat B–1 toward Cat 3 / PL d — a safety relay or safety-rated PLC, dual-channel inputs, two contactors (or force-guided contacts) with feedback (EDM) monitoring, and a synchronous two-hand module per ISO 13851.

- **Ferrule the control wiring:** crimp ferrules on stranded control conductors at the terminal blocks to eliminate the loose-strand faults seen during commissioning.

13.2 Monitoring and Connectivity

- **Telemetry:** stream live operating data (motor current, speed, jam count, fault state, runtime) off the PLC for dashboards and trend analysis.
- **Modbus to the VFD:** the Huanyang supports RS-485 Modbus (currently unused). Adding it would pull far richer data straight from the drive — output current, voltage, frequency/RPM, DC-bus voltage, drive temperature, and detailed fault codes — beyond the single 0–10 V analog monitor.
- **Internet / remote viewing:** expose a read-only view of device status over the network — via the H2-DM1E's built-in web server or an ESP32 / edge gateway — so the shredder can be monitored remotely from a browser or phone.
- **Data logging:** record total runtime, jam events per shift, and faults locally for predictive maintenance.

13.3 Capability

- **Predictive jam detection:** train a small model on the motor-current waveform shape just before a jam to detect and pre-empt jams earlier.
- **Mechanical integration:** add a feed hopper that auto-meters plastic into the blades.
- **Higher-capacity model:** scale to a 5–10 HP drivetrain for greater throughput.

14 Ethics and Design Considerations

A summary of the ethics and design evaluation follows:

- **Public health and safety:** Yes – layered safety functions (hardwired E-stop, two-hand deadman control), NFPA 70 / NEMA compliance, hardwired safety chain.
- **Environmental:** Yes – core project purpose is diverting plastic from landfill.
- **Economic:** Yes – significant cost savings versus commercial alternatives, careful component selection within budget.
- **Global:** Possibly – approach is replicable globally, components internationally sourced.
- **Cultural:** No – industrial machine for B2B use, no culture-specific design.

15 References

Codes and Standards

- NFPA 70 (2023). *National Electrical Code* (Articles 110, 250, 310, 409, 430). National Fire Protection Association, Quincy, MA.
  nfpa.org/codes-and-standards (NFPA 70)
- NFPA 79 (2024). *Electrical Standard for Industrial Machinery.* NFPA, Quincy, MA.
  nfpa.org/codes-and-standards (NFPA 79)
- NEMA 250 (2020). *Enclosures for Electrical Equipment (1000 Volts Maximum).* National Electrical Manufacturers Association.
  nema.org (NEMA 250)
- NEMA ICS 1 (2000, R2020). *Industrial Control and Systems: General Requirements.*
  nema.org (NEMA ICS 1)
- NEMA ICS 2 (2000, R2020). *Industrial Control and Systems: Controllers, Contactors, and Overload Relays Rated 600 Volts.*
  nema.org (NEMA ICS 2)
- NEMA MG 1 (2021). *Motors and Generators.*
  nema.org (NEMA MG 1)
- UL 508A (2018). *Standard for Industrial Control Panels.* Underwriters Laboratories.
  shopulstandards.com (UL 508A)
- IEC 60204-1:2016. *Safety of Machinery: Electrical Equipment of Machines, Part 1.* International Electrotechnical Commission.
  webstore.iec.ch/publication/26037
- ISO 12100:2010. *Safety of Machinery: General Principles for Design, Risk Assessment and Risk Reduction.* International Organization for Standardization.
  iso.org/standard/51528.html
- ISO 13849-1:2023. *Safety of Machinery: Safety-related Parts of Control Systems, Part 1.* ISO.
  iso.org/standard/73481.html
- ISO 13850:2015. *Safety of Machinery: Emergency Stop Function.* ISO.
  iso.org/standard/59970.html
- ISO 13851:2019. *Safety of Machinery: Two-Hand Control Devices.* ISO.
  iso.org/standard/70295.html

- OSHA 29 CFR 1910.147. *The Control of Hazardous Energy (Lockout/Tagout).* U.S. Department of Labor, Occupational Safety and Health Administration.
osha.gov/laws-regs/regulations/standardnumber/1910/1910.147

Manufacturer Datasheets and Manuals

- Huanyang Electrical, *HY-series Variable Frequency Drive User Manual* (HY02D211B-T).
huanyanginverter.com
- AutomationDirect, *DL205 User Manual* (D2-USER-M).
automationdirect.com (D2-USER-M)
- AutomationDirect, *Do-more H2 Series CPU User Manual* (HX-USER-M).
automationdirect.com (HX-USER-M)
- AutomationDirect, *C-more Micro-Graphic Panels User Manual* (EA-MG-USER-M).
automationdirect.com (EA-MG-USER-M)
- Mean Well, *NDR-480 Series 480 W Single Output Industrial DIN Rail Power Supply Datasheet.*
meanwell.com (NDR-480)
- Mitsubishi Electric, *SD-N Series Magnetic Contactors Catalog* (SD-N35).
mitsubishielectric.com (SD-N magnetic contactors)
- ATO, *APCS-300R30-AD 300 W 30 Ω Brake Resistor Datasheet.*
ato.com (300 W 30 Ω braking resistor)
- General Electric, *5KE Frame 182T Motor Nameplate / Service Data* (5KE182BC205B).
Archived service literature.
- PowerTec, *71008 Two-Pole START/STOP Switch Product Page.*
powertecmotors.com
- Cutler-Hammer (Eaton), *E22 Series 22 mm Pushbutton Operator Catalog* (E22B1 mushroom E-stop).
eaton.com (E22 series)
- Vorksmen, *SEB001 Industrial Steel Control Panel Enclosure Product Specification.* NEMA Type 12, gasketed steel enclosure used as the main panel housing. Manufacturer-rated rating sheet on file with the team; sourcing entry in Appendix A (BOM).

Course and Sponsor Materials

- ECE 412/413 Senior Capstone Project Development I/II course materials and project proposal, Portland State University, Maseeh College of Engineering and Computer Science, Winter and Spring 2026.

- MME / Ichor Systems, *Universal Plastic Shredder V2.0 Sponsor Brief*, Bridget Lannigan, 2026 (internal communications, archived in repository).
- PSU Power Engineering Laboratory, *EE 347L Switchgear Project Final Report Template* (basis for the document structure and PSU MCECS report style used here).

16 Appendices

16.1 Appendix A: Bill of Materials

The complete as-built Bill of Materials is consolidated in `BOM/BOM_Combined_As_Built.xlsx` (and the rendered `BOM/BOM_Combined_As_Built.pdf`) in the project repository. The workbook contains three sheets:

- **Combined BOM** — every item actually installed in the control panel, grouped by category (Motor & Drive, Power Distribution, PLC Control System, HMI, Operator Controls, Motor Connections, Wire/Duct/Hardware) with manufacturer, part number, quantity, and notes pointing back to the relevant documentation.
- **Excluded - Not Used** — items that appeared in preliminary BOMs but were not used in the final build, with the reason (rejected during selection, not purchased, superseded by another component, or out of project budget).
- **VFD Decision Matrix** — the weighted comparison used to select the Huanyang HY02D211B-T (price 50%, current capability 20%, build quality 15%, documentation 10%, brand 5%).

Critical items (highlighted in the workbook): Motor (GE 5KE182BC205B for bench test), VFD (Huanyang HY02D211B-T), Brake Resistor (ATO APCS-300R30-AD), Mean Well NDR-480-24 24V DC power supply, Mitsubishi SD-N35 contactor, Cutler-Hammer E22B1 E-stop, AutomationDirect PLC rack with H2-DM1E CPU and the D2-08ND3 / F2-08AD-1 / F2-04RTD / D2-08TR / F2-08DA-2 modules, EA1-T4CL HMI.

Two motors appear in this build. The production motor on the as-built panel is the IronHorse MTCP2-002 (2 HP, 230 V, 5.93 A FLA, 4-pole, 1735 RPM, NEMA 145TC TEFC IP55, VFD-rated 1.0 SF); the GE 5KE182BC205B (3 HP, 230 V, 7.6 A, 4-pole, 1750 RPM) was the bench/test motor used to validate the control system through most of the term. The VFD parameter set in the panel today still reflects the GE bench values pending re-program with the IronHorse nameplate (`PD141 = 230`, `PD142 = 5.93`, `PD143 = 4`, `PD144 = 1735`); see the open-work punch list `docs/term2_week11_Open_Work.md` item 4. Control-panel detail (rack layout, terminal-by-terminal wiring) is itemized in the point list in Appendix B.

The rendered workbook follows on the landscape pages below.

PRODUCT NAME	Shredder	**EE TEAM**	
APPROVED BY		**Fearghus Tyler**	
DATE OF APPROVAL	2026-06-01	**Bao Nguyen**	
PART COUNT	62	**Yaqoub Rabiah**	
TOTAL COST	$969.88	**Fox Kang**	

PART NUMBER	PART NAME	DESCRIPTION	QUANTITY	UNITS	LINK	UNIT COST	AMOUNT
5KE182BC205B	AC Motor (3 HP)	GE 3-phase TEFC induction motor; 230 V (low-voltage), 1750 RPM, 7.6 A FLA, NEMA 182T frame. Bench-test motor.	1	EA	https://www.electricmotorsforless.com/	$0.00	$0.00
HY02D211B-T	VFD	Huanyang variable frequency drive, 110 V single-phase input, 220 V 3-phase output, 2.2 kW (3 HP). Selected via decision matrix.	1	EA	https://www.amazon.com/Huanyang-1-Phase-Variable-Frequency-Output-3hp/dp/B0DMF3GB38	$200.00	$200.00
APCS-300R30-AD	Brake Resistor	LS Electric 300 W, 30 ohm aluminum-clad braking resistor for VFD dynamic braking on P+/PR. Purchased from AutomationDirect. Note: an earlier identical-part-number unit (ATO branded) was ordered but never arrived after shipping; replaced with this LS Electric unit which is the brake resistor actually installed.	1	EA	https://www.automationdirect.com/adc/shopping/catalog/motion_control/ls_electric_l7c_(l7ca_series)_servo_systems/servo_accessories/apcs-300r30-ad	$29.00	$29.00
71008	Main Power Switch	Powertec 120 V ON/OFF main disconnect switch for entire control panel.	1	EA	https://powertecproducts.com/71008/	$15.00	$15.00
TMC 8 3C 15A (2907618)	VFD Branch Breaker	Phoenix Contact thermal-magnetic DIN-rail circuit breaker, 15 A, on VFD neutral leg (per as-built). Obtained at no cost from the PSU EPL free pile.	1	EA	https://www.phoenixcontact.com/	$0.00	$0.00
NDR-480-24	24 VDC Power Supply	Mean Well 480 W, 24 VDC DIN-rail power supply for PLC, HMI, indicator LEDs, and VFD control loops. Donated by sponsor / obtained at no cost to the EE team.	1	EA	https://www.meanwell.com/	$0.00	$0.00
SD-N35	Main Power Contactor	Mitsubishi 3-pole magnetic contactor. Energized through E-stop chain via E22B1 NC contact.	1	EA	https://www.ebay.com/itm/... (tpmachinery seller, order 23-14145-39817, Jan 2026)	$12.59	$12.59
EURO S4LH	Fuse Block (button common)	Morsettitalia fuse-holder terminal block. Holds 1 A fast-blow fuse protecting the shared pushbutton +24 V supply.	1	EA	https://www.morsettitalia.com/	$8.00	$8.00
EURO S10H-5H	Fuse Block (AC branches)	Morsettitalia fuse-holder terminal block for AC branch protection.	1	EA	https://www.morsettitalia.com/	$12.00	$12.00
H2-DM1E	PLC CPU / Ethernet	AutomationDirect Do-more CPU with built-in Ethernet, fits D2-09B-1 DL205 rack. Purchased Jan 2026 from planet-surplus as a USED combo for $350 total, includes: D2-09B-1 chassis, D2-08ND3, F2-08AD-1, F2-04RTD, D2-08TR, F2-08DA-2, and D2-FILL filler panels.	1	EA	https://www.ebay.com/itm/... (planet-surplus seller, order 20-14113-87209, Jan 2026)	$350.00	$350.00
D2-09B-1	PLC Base / Rack	AutomationDirect DirectLOGIC 205 9-slot AC-powered base. Included in the H2-DM1E used combo purchase.	1	EA	https://www.automationdirect.com/	$0.00	$0.00
D2-08ND3	Discrete Input Module	AutomationDirect 8-ch, 24 VDC, sourcing/sinking discrete input module. Slot 2. Inputs X0-X7. Included in the H2-DM1E used combo purchase.	1	EA	https://www.automationdirect.com/	$0.00	$0.00
F2-08AD-1	Analog Input Module	AutomationDirect 8-ch, 4-20 mA analog current input module. Slot 3. CH1+ reads VFD VO current monitor. Included in the H2-DM1E used combo purchase.	1	EA	https://www.automationdirect.com/	$0.00	$0.00
F2-04RTD	RTD Input Module	AutomationDirect 4-ch RTD input module. Slot 4. Installed but unused (spare). Included in the H2-DM1E used combo purchase.	1	EA	https://www.automationdirect.com/	$0.00	$0.00
D2-08TR	Relay Output Module	AutomationDirect 8-ch relay output module. Slot 5. Drives button LEDs (Y0-Y4) and VFD FOR/REV/RST (Y6/Y7/Y5). Included in the H2-DM1E used combo purchase (eBay listing showed D2-08TD2 but seller substituted D2-08TR which is what we use).	1	EA	https://www.automationdirect.com/	$0.00	$0.00
F2-08DA-2	Analog Output Module	AutomationDirect 8-ch, 0-10 V analog voltage output module. Slot 6. +V1 drives VFD VI speed reference. Included in the H2-DM1E used combo purchase (eBay listing showed F2-02DA but actual unit is F2-08DA-2).	1	EA	https://www.automationdirect.com/	$0.00	$0.00

PART NUMBER	PART NAME	DESCRIPTION	QUANTITY	UNITS	LINK	UNIT COST	AMOUNT
D2-FILL	Rack Filler Panels	AutomationDirect blank fill panels for unused rack slots 7-9. Included in the H2-DM1E used combo purchase.	3	EA	https://www.automationdirect.com/	$0.00	$0.00
EA1-T4CL	HMI Touch Panel	AutomationDirect C-More Micro-Graphic 4-inch color touch panel HMI. Purchased used from eBay for $148.95 (Apr 2026). Purchase included the serial cable for HMI-to-CPU communication (RS-232 to H2-DM1E serial port).	1	EA	https://www.ebay.com/itm/306662618920	$148.95	$148.95
S2PR-P3G	Forward Pushbutton Head	Autonics 22 mm illuminated green pushbutton head (Forward command -> X0).	1	EA	https://www.autonics.com/us/model/S2PR-P3GAD	$1.90	$1.90
S2PR-P3Y	Reverse Pushbutton Head	Autonics 22 mm illuminated yellow pushbutton head (Reverse command -> X1).	1	EA	https://www.autonics.com/us/model/S2PR-P3YAD	$1.90	$1.90
S2PR-P3B	Run/Deadman Pushbutton Head	Autonics 22 mm illuminated blue pushbutton head (Run 1 -> X2, Run 2 -> X3).	2	EA	https://www.autonics.com/us/model/S2PR-P3BAD	$1.90	$3.80
S2PR-P3R	Reset Pushbutton Head	Autonics 22 mm illuminated red pushbutton head (planned Reset -> X6).	1	EA	https://www.autonics.com/us/model/S2PR-P3RAD	$4.50	$4.50
SA-CA	NO Contact Block (1NO)	Autonics finger-safe 1NO contact block. 110 VAC/10 A. Used with each S2PR pushbutton.	5	EA	https://www.autonics.com/us/model/SA-CA	$2.80	$14.00
SA-LDG	LED Block - Green (24 VDC)	Autonics modular LED indicator block, green, 12-24 VDC. Forward button illumination -> Y0.	1	EA	https://www.autonics.com/us/model/SA-LDG	$4.50	$4.50
SA-LDY	LED Block - Yellow (24 VDC)	Autonics modular LED indicator block, yellow, 12-24 VDC. Reverse button illumination -> Y1.	1	EA	https://www.autonics.com/us/model/SA-LDY	$4.50	$4.50
SA-LDB	LED Block - Blue (24 VDC)	Autonics modular LED indicator block, blue, 12-24 VDC. Run 1 and Run 2 illumination -> Y2, Y3.	2	EA	https://www.autonics.com/us/model/SA-LDB	$4.50	$9.00
SA-LDR	LED Block - Red (24 VDC)	Autonics modular LED indicator block, red, 12-24 VDC. Reset button illumination (planned).	1	EA	https://www.autonics.com/us/model/SA-LDR	$4.50	$4.50
E22B1	Emergency Stop Pushbutton	Cutler-Hammer / Eaton 22 mm mushroom E-stop, NC contact. In series with SD-N35 contactor coil.	1	EA	https://www.eaton.com/	$25.00	$25.00
Wamco 525	Fault Indicator Light	Wamco panel-mount LED indicator, 28 VDC, 0.6 W. Driven by PLC Y4 as machine-fault indicator.	1	EA	https://www.wamco.com/	$15.00	$15.00
Banana Socket (Blue)	Banana Socket Blue	Generic 4 mm panel-mount banana socket, blue. Quick-disconnect for VFD U output to motor T1. Obtained at no cost from a free pile.	1	EA		$0.00	$0.00
Banana Socket (White)	Banana Socket White	Generic 4 mm panel-mount banana socket, white. VFD V output to motor T2. Obtained at no cost from a free pile.	1	EA		$0.00	$0.00
Banana Socket (Red)	Banana Socket Red	Generic 4 mm panel-mount banana socket, red. VFD W output to motor T3. Obtained at no cost from a free pile.	1	EA		$0.00	$0.00
G1X1LG6	Slotted Wire Duct, 1"x1"	Panduit Type G wide-finger slotted PVC duct, light gray, 6 ft lengths.	6	FT	https://www.panduit.com/	$3.43	$20.58
C1LG6	Duct Cover, 1"W	Panduit duct cover for G1X1LG6, light gray, 6 ft lengths.	6	FT	https://www.panduit.com/	$1.20	$7.20
18 AWG Blue	Control Wire, Blue	18 AWG stranded blue control wire (NFPA 79 §13.2 DC control). Obtained at no cost (some from team member's existing stock, some from free pile).	1	Roll		$0.00	$0.00
18 AWG White	Control Return Wire	18 AWG stranded white wire for DC return (NFPA 79 §13.2). Obtained at no cost (some from team member's existing stock, some from free pile).	1	Roll		$0.00	$0.00
12 AWG Black	Power Wire, Black	12 AWG black wire for AC mains and motor leads. Obtained at no cost (some from team member's existing stock, some from free pile).	1	Roll		$0.00	$0.00
12 AWG Green/Yellow	Ground Wire	12 AWG green/yellow stranded equipment grounding wire (NFPA 79). Obtained at no cost (some from team member's existing stock, some from free pile).	1	Roll		$0.00	$0.00
GMA-500mA or 0217.500MXP	Fuse - 500 mA F	Bussmann or Littelfuse 5x20 mm fast-blow fuse, 500 mA, 250 V. (Currently not installed per as-built note - retained as recommended addition.)	1	EA		$1.50	$1.50
GMA-1A or equiv.	Fuse - 1 A F	5x20 mm fast-blow fuse, 1 A, 250 V. Protects shared +24 V pushbutton common (in EURO S4LH).	1	EA		$1.50	$1.50
LM2596	DC-DC Buck Converter (24V to 5V/3.3V)	Seloky LM2596-based adjustable DC-DC buck converter module with voltmeter display (sold as 2-pack). Steps down 24 VDC panel supply to the input voltage required by the capacitive touch module.	1	Pack	https://www.amazon.com/Converter-Adjustable-Regulator-Voltmeter-Display/dp/B0F1M2SR31	$7.99	$7.99

PART NUMBER	PART NAME	DESCRIPTION	QUANTITY	UNITS	LINK	UNIT COST	AMOUNT
SEB001	Control Panel Enclosure	Vorksmen industrial control enclosure (steel) housing the entire electrical system including PLC rack, VFD, contactor, PSU, terminal blocks, and operator panel.	1	EA	https://www.amazon.com/	$59.98	$59.98
AE14921a	3V Relay	AEDIKO 3 VDC PCB-mount relay. Used in the capacitive touch module sub-circuit to interface the low-voltage touch sensor output with the 24 VDC panel control voltage.	1	EA	https://www.amazon.com/	$6.99	$6.99
DGN-001	*Bao's Dignity*	*Lost on the procurement loop after lowballing roughly every vendor between Portland and Beaverton. Quoted "open-box" / "educational discount" / "the next one is on us" until even PowerTec stopped replying. Non-returnable. Replacement TBD; budget is hopefully zero.*	1	EA		$0.00	$0.00
		TOTAL PARTS	62		TOTAL		$969.88

PART NUMBER	PART NAME	DESCRIPTION	FROM BOM FILE	REASON NOT USED
PE145TC-2-4	Motor (alternate)	Original BOM candidate motor	BOM_02_20_26	Not used; GE 5KE182BC205B used instead for bench testing
NBE7	Circuit Breaker	Original BOM candidate breaker	BOM_1_29 / BOM_02_20_26	Never purchased; replaced by Phoenix Contact TMC 8 3C 15A
(30 A fuse, TBD)	30 A Branch Fuse	Per old BOM, 2x 30 A fuses budgeted	BOM_1_29	Never needed in final design
1866-1441-ND	DC Power Supply	120/24 VDC PSU from Digi-Key	BOM_1_29	Superseded by Mean Well NDR-480-24
(generic)	VEVOR 3HP 120V VFD	Hobby-grade VFD option	VFD_120V_3HP_Decision_Matrix	Rejected during selection (hobby-grade, limited documentation)
(generic)	ATO 3HP 120V VFD	Higher-quality VFD option	VFD_120V_3HP_Decision_Matrix	Too expensive ($450-500) for project budget
(generic)	110V 20A 3HP VFD	Mid-tier VFD option	VFD_120V_3HP_Decision_Matrix	Huanyang HY02D211B-T selected instead
(WEG single-phase 3HP)	WEG Single Phase Motor	1-phase general purpose 3 HP motor	Motor Spec	Considered but GE 5KE182BC205B used
NT3-18-213T	Worldwide Electric Motor	3-phase TEFC 3 HP motor	Motor Spec	Considered but not purchased
(VEVOR 5HP 230V)	VEVOR High Torque Motor	5 HP 1-phase compressor-style motor	Motor Spec	Considered but not purchased
L510 (2HP / 3HP)	TECO L510 VFD	VFD comparison option	Motor Spec	Considered but Huanyang selected
G0532	Marathon Gearmotor	Integrated gearbox motor	Motor Spec	Considered but not purchased
KBWC-15K	KB Electronics speed control	Triac speed control fan-style	Motor Spec	VFD approach used instead
TTP223B	Capacitive Touch Sensor	Single-channel cap touch	BOM_1_30_beta (safety draft)	Does not meet industrial standards
CAP1293-1-SN	Capacitive Touch Chip	3-channel cap touch chip	BOM_1_30_beta (safety draft)	Chip only; needs interface board
1864-1032-ND	Capacitive Proximity Sensor	Cap proximity sensor	BOM_1_30_beta (safety draft)	Too expensive ($109.25)
F3S-TGR-NMPC-21-10	Omron Magnetic Safety Sensor	Non-contact magnetic safety switch	BOM_1_30_beta (safety draft)	Too expensive; out of stock
RSS 36-I2-D-R-ST5	Schmersal RFID Safety Sensor	RFID safety sensor	BOM_1_30_beta (safety draft)	Suspected industrial-only / non-retail
F3SG-SR / PG	Omron Safety Light Curtain	Safety light curtain	BOM_1_30_beta (safety draft)	Out of scope for capstone budget
(Ichor-provided)	Red/Green Indicator Lights	Sponsor-provided lights	BOM_02_20_26	Not used; Wamco 525 fault light and button LEDs cover indication
(Ichor-provided)	E-stop Pushbutton	Sponsor-provided E-stop	BOM_02_20_26	Replaced with Cutler-Hammer/Eaton E22B1 procured by team
V30AE000055	Insulated Ferrules, 12 AWG (gray)	Z+F crimp ferrules, 12 AWG, 10 mm length, gray (DIN color code). 500-piece pack.	URS Electronics invoice (purchased but not used)	Purchased from URS Electronics but ultimately not used in the as-built panel. Spare stock retained for future projects.

16.2 Appendix B: PLC I/O Point List

The complete as-built point list, transcribed from the project workbook (`Point_List.xlsx`), follows on the landscape pages below — every sheet, using the spreadsheet's columns *Point Description, Origin Address, DO / DI / AO / AI / Pwr, Destination Address, Destination Description, and Notes*: the index; the Mitsubishi SD-N35 contactor; the Mean Well NDR-480-24 power supply; the DirectLOGIC 205 rack and the H2-DM1E CPU; the EA1-T4CL HMI; every I/O module (D2-08ND3, F2-08AD-1, F2-04RTD, D2-08TR, F2-08DA-2), the legacy D2-08TA and F2-02DAS-2 sheets; and the Huanyang HY02D211B-T VFD.

Sheet: Index

Sheet #	Sheet Name
1	Mitsubishi SD-N35
2	Mean Well NDR-480-24
3	Direct Logic 205 D2-09B-1
4	DirectLOGIC CPU (H2-DM1E)
5	Output Module (D2-08TA)
6	HMI (EA1-T4CL)
7	Output Module (F2-02DAS-2)
8	Input Module (D2-08ND3)
9	Analog Input (F2-08AD-1)
10	RTD Input (F2-04RTD)
11	Relay Output (D2-08TR)
12	Analog Output (F2-08DA-2)
13	VFD (Huanyang HY02D211B-T)

Sheet: Index — Revision History

	REV	DATE	DESCRIPTION	Created By	Checked by
	1	2026-03-14	Initial point list (end of Winter term).	Bao Nguyen	Fearghus Tyler
	2	2026-05-18	Updated with as-built wiring and module corrections.	Bao Nguyen	Fearghus Tyler
	3	2026-06-01	Added missing component sheets to index; full revision log.	Bao Nguyen	Fearghus Tyler

Sheet: Mitsubishi SD-N35

Point Description	**Origin**	**DO**	**DI**	**AO**	**AI**	**Pwr**	**Destination Address**	**Destination Description**	**Notes**
Contactor (Mitsubishi SD-N35) – Coil A1	A1	0	0	0	0	1	Y0	D2-08TD2 Output Module (12–24VDC)	A1 → DF103V (1/2A slow blow fuse) → D2-08TD2 output supplying +24VDC
Contactor (Mitsubishi SD-N35) – Coil A2	A2	0	0	0	0	1	-V	NDR-480-24 Power Supply	A2 → direct return to power supply negative (–V)
Contactor (Mitsubishi SD-N35) – Line Input	L1	0	0	0	0	1	L	Phoenix Contact TMC 83C 15A (2907618)	L1 → Phoenix Contact fuse breaker → fed from wall AC line
Contactor (Mitsubishi SD-N35) – Load Output	T1	0	0	0	0	1	R	HY Series VFD (HY02D211B-T)	T1 → passes through contactor → supplies VFD terminal R (line input)
Total Points		**0**	**0**	**0**	**0**	**4**			

Sheet: Mean Well NDR-480-24

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
		0					AC Mains Hot from wall via main power switch and 25 A fuse	120 VAC line conductor from wall outlet, routed through panel main disconnect, then through 25 A slow-blow fuse on EURO S10H-5H fuse block	Wall AC L -> main disconnect -> EURO S10H-5H 25 A slow-blow -> PSU L. Black wire per NFPA 79.
Mean Well NDR-480-24 N (AC Neutral Input)	N	0	0	0	0	1		120 VAC neutral conductor from wall outlet, returns through neutral bus	Wall AC N -> neutral bus -> PSU N. White wire per NFPA 79.
Mean Well NDR-480-24 G (Ground / PE)	G	0	0	0	0	1	Panel Ground Bus	Equipment grounding conductor; bonds PSU chassis to panel ground bus and ultimately to building earth	PSU G stud -> panel ground bus. Green or Green/Yellow wire per NFPA 79.
Mean Well NDR-480-24 +V to EURO S4LH (Pushbutton Common)	+V	0	0	0	0	1	EURO S4LH Terminal Block -> Pushbutton Commons	Fused +24V distribution to operator pushbutton common via S4LH	+V -> 1 A fast-blow fuse (EURO S4LH terminal) -> distributes to common side of all illuminated pushbuttons (Forward S2PR-P3G, Reverse S2PR-P3Y, Run 1/2 S2PR-P3B x2, Reset S2PR-P3R).
Mean Well NDR-480-24 +V to E22B1 E-Stop Circuit	+V	0	0	0	0	1	Cutler-Hammer E22B1 NC contact + SD-N35 contactor coil A1	Direct +24V (no fuse) into E-stop safety chain	+V -> E22B1 NC contact -> SD-N35 A1 (contactor coil). PLC X4 taps this line between E22B1 and A1 to monitor E-stop state.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Mean Well NDR-480-24 +V to VFD FA (Jam Signal)	+V	0	0	0	0	1	VFD Huanyang HY02D211B-T FA terminal	Direct +24V (no fuse) into VFD jam signal circuit	+V -> VFD FA. When VFD fault relay closes (PD052=02), FA-FC bridges and delivers +24V to PLC X5.
Mean Well NDR-480-24 +V to F2-08AD-1 (Module Power)	+V	0	0	0	0	1	F2-08AD-1 Analog Input Module +24V terminal	Direct module power	+V -> F2-08AD-1 +24V terminal. No fuse on this branch.
Mean Well NDR-480-24 +V to F2-08DA-2 (Module Power)	+V	0	0	0	0	1	F2-08DA-2 Analog Output Module +24V terminal	Direct module power	+V -> F2-08DA-2 +24V terminal. No fuse on this branch.
Mean Well NDR-480-24 +V to F2-04RTD (Module Power)	+V	0	0	0	0	1	F2-04RTD RTD Input Module +24V terminal	Direct module power	+V -> F2-04RTD +24V terminal. No fuse on this branch.
Mean Well NDR-480-24 +V to Indicator LEDs	+V	0	0	0	0	1	LED indicator (+) terminals (SA-LDG/Y/B/B/R, Wamco 525)	Direct +24V to (+) side of all illuminated push-button LED blocks and the separate Wamco error light	+V -> SA-LDG (+), SA-LDY (+), SA-LDB (+) x2, SA-LDR (+) (when wired), Wamco 525 (+). LED (-) sides return to D2-08TR relay outputs Y0-Y4 which sink to -V when energized.
Mean Well NDR-480-24 -V to SD-N35 Contactor Coil A2	-V	0	0	0	0	1	Mitsubishi SD-N35 Contactor A2 (coil return)	Contactor coil return through E-stop chain	-V -> SD-N35 A2 terminal. Coil energizes when E22B1 NC closes (E-stop not pressed) and +V reaches A1.
Mean Well NDR-480-24 -V to D2-08ND3 (Input Common)	-V	0	0	0	0	1	D2-08ND3 Discrete Input Module Common (C)	Module input common for sourcing mode operation	-V -> D2-08ND3 C terminal. Sourcing mode reference - X inputs read HIGH when +24V is applied.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Mean Well NDR-480-24 -V to D2-08TR (Relay Common)	-V	0	0	0	0	1	D2-08TR Relay Output Module Common (C)	Relay common for sinking mode operation	-V -> D2-08TR C terminal. When any Y0-Y7 relay closes, that output terminal connects to -V (sinks current from the load).
Mean Well NDR-480-24 -V to F2-08AD-1 (Module Power Return)	-V	0	0	0	0	1	F2-08AD-1 Analog Input Module 0V terminal	Module power return and analog signal common	-V -> F2-08AD-1 0V terminal.
Mean Well NDR-480-24 -V to F2-08DA-2 (Module Power Return / Analog Common)	-V	0	0	0	0	1	F2-08DA-2 Analog Output Module 0V terminal	Module power return and analog signal reference for all 8 channels	-V -> F2-08DA-2 0V terminal. Single 0V terminal also serves as the analog signal common.
Mean Well NDR-480-24 -V to F2-04RTD (Module Power Return)	-V	0	0	0	0	1	F2-04RTD RTD Input Module 0V terminal	Module power return	-V -> F2-04RTD 0V terminal.
Mean Well NDR-480-24 -V to VFD DCM (Digital Common)	-V	0	0	0	0	1	VFD Huanyang HY02D211B-T DCM terminal	VFD digital input common reference	-V -> VFD DCM. Provides 0V reference for FOR/REV/RST inputs (sinking/NPN mode). Closes the loop when D2-08TR relays Y5/Y6/Y7 pull those VFD inputs to DCM.
Mean Well NDR-480-24 -V to VFD ACM (Analog Common)	-V	0	0	0	0	1	VFD Huanyang HY02D211B-T ACM terminal	VFD analog signal common reference	-V -> VFD ACM. Shared 0V reference for the analog speed command (VI from F2-08DA-2) and the analog current monitor (VO to F2-08AD-1).
Mean Well NDR-480-24 +V to HMI (EA1-T4CL)	+V	0	0	0	0	1	C-More EA1-T4CL HMI +VDC input	HMI panel power	+V -> HMI +VDC. 24 VDC supply for the C-More Micro-Graphic panel (replaces former Rhino PSB12-030-P).

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Mean Well NDR-480-24 -V to HMI (EA1-T4CL)	-V	0	0	0	0	1	C-More EA1-T4CL HMI 0VDC return	HMI panel power return	-V -> HMI 0VDC. System DC common reference for the HMI panel.
Total Points		**0**	**0**	**0**	**0**	**19**			

Sheet: Direct Logic 205 D2-09B-1

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
PLC Rack (DirectLOGIC 205) – Base	Base	0	0	0	0	1	D2-09B-1	Base / Power Unit	Rack base unit providing backplane power and I/O bus to all installed modules. Supports up to 9 I/O slots.
PLC Rack (DirectLOGIC 205) – Line Input	L	0	0	0	0	1	L	MORSETTITALIA EURO S10H-5H Fuse Block	Line from wall -> passes through power switch (closed = line passes) -> enters EURO S10H-5H fuse block with 1A fast blow fuse (PLC branch) -> enters PLC L terminal
PLC Rack (DirectLOGIC 205) – Neutral Input	N	0	0	0	0	1	N	Neutral Bus	Neutral from wall -> directly connected to PLC N terminal
PLC Rack (DirectLOGIC 205) – Ground (Protective Earth)	G	0	0	0	0	1	G	Ground Bus	Protective earth -> connected to PLC ground terminal / chassis bond
PLC Rack (DirectLOGIC 205) – Slot 1	Slot 1	0	0	0	0	0	H2-DM1E	CPU / Ethernet Module	CPU module with built-in Ethernet port. Executes the ladder logic program and handles network communications.
PLC Rack (DirectLOGIC 205) – Slot 2	Slot 2	0	8	0	0	0	D2-08ND3	8-Point Discrete Input Module	8-channel 24VDC discrete (digital) input module. Accepts sink/source wiring. Used for reading on/off signals such as pushbuttons, limit switches, and proximity sensors.
PLC Rack (DirectLOGIC 205) – Slot 3	Slot 3	0	0	0	8	0	F2-08AD-1	8-Channel Analog Current Input Module (4-20 mA)	8-channel current input module. Used for motor current monitoring (CH1+ to VFD VO terminal, scaled 0-10V from VFD).

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
PLC Rack (DirectLOGIC 205) – Slot 4	Slot 4	0	0	0	4	0	F2-04RTD	4-Channel RTD Input Module	4-channel RTD (Resistance Temperature Detector) input module. Compatible with PT100/PT1000 and other RTD types. Used for precision temperature measurement.
PLC Rack (DirectLOGIC 205) – Slot 5	Slot 5	8	0	0	0	0	D2-08TR	8-Point Relay Output Module	8-channel relay (dry contact) output module. Common to PSU -V (sinking mode). Drives indicator LEDs (Y0-Y4) and VFD digital inputs (Y5=RST, Y6=FOR, Y7=REV).
PLC Rack (DirectLOGIC 205) – Slot 6	Slot 6	0	0	8	0	0	F2-08DA-2	8-Channel Analog Voltage Output Module (0-10 V)	8-channel voltage output module. CH1 (+V1) provides VFD speed reference to VI terminal (0-10V scaled to frequency).
PLC Rack (DirectLOGIC 205) – Slot 7	Slot 7	—	—	—	—	—	D2-Fill	Fill Panel	Blank fill panel. No I/O. Slot reserved for future expansion.
PLC Rack (DirectLOGIC 205) – Slot 8	Slot 8	—	—	—	—	—	D2-Fill	Fill Panel	Blank fill panel. No I/O. Slot reserved for future expansion.
PLC Rack (DirectLOGIC 205) – Slot 9	Slot 9	—	—	—	—	—	D2-Fill	Fill Panel	Blank fill panel. No I/O. Slot reserved for future expansion.
Total Points		**8**	**8**	**8**	**12**	**4**			

Sheet: DirectLOGIC CPU (H2-DM1E)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Power and communication interface	Backplane Connector (Left Edge)	0	0	0	0	1	D2-09B-1 Base Slot 1 Connector	Provides power and backplane communication	–
Ground reference	Backplane GND	0	0	0	0	1	D2-09B-1 Base Ground Bus	Common ground for logic power	–
HMI communication port	CPU Comm Port	0	0	0	0	0	HMI EA1-T4CL COM Port	4” C-More Micro-Graphic panel (operator interface)	CPU communication port links to the C-More HMI (see HMI sheet). Match baud, parity, and stop bits on both ends. Carries operator commands to the PLC and status/data back to the panel.
Total Points		**0**	**0**	**0**	**0**	**2**			

Sheet: Output Module (D2-08TA)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Output Common (Hot)	C	1	0	0	0	1	120 V Hot Bus	ZipLink Terminal Block (ZL-RTB20-OUT). 1A in-line slow blow fuse	120 V hot feed shared by all outputs
Output 0	OUT0	1	0	0	0	0	A1	TX Contactor (GH15-BN)	Drives TX coil via ZipLink
Output 1	OUT1	1	0	0	0	0	A1	BUS Contactor (GH15-BN)	Drives BUS coil via ZipLink
Output 2	OUT2	1	0	0	0	0	A1	F1 Contactor (GH15-BN)	Drives F1 coil via ZipLink
Output 3	OUT3	1	0	0	0	0	95	Thermal Overload Relay (RTD32-180)	Feeds NC contact 95 $\rightarrow$ 96 $\rightarrow$ F2 coil
Module Power Feed	Backplane	0	0	0	0	1	Slot 3 Connector	PLC base backplane (120 VAC)	No external wiring
Total Points		**5**	**0**	**0**	**0**	**2**			

Sheet: HMI (EA1-T4CL)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
HMI Serial Comm Port	COM Port	0	0	0	0	0	Serial Port	H2-DM1E	Serial link between PLC CPU and C-More HMI. Configure PLC and HMI ports for matching baud, parity, stop bits
HMI Power +V	+VDC	0	0	0	0	1	+V	Mean Well NDR-480-24	HMI 24 VDC input powered from Mean Well NDR-480-24 +V rail
HMI Power 0 V	0VDC	0	0	0	0	1	0V	Mean Well NDR-480-24	0 VDC return to Mean Well NDR-480-24 -V rail (system DC common)
HMI Ground	Ground	0	0	0	0	1	Ground	Ground	
Total Points		**0**	**0**	**0**	**0**	**3**			

Sheet: Output Module (F2-02DAS-2)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Analog Common (CH1)	CH1-V	0	0	1	0	0	CM	VFD (GS1-20P2) through Ziplink ZL-RTB20	0V1 and CH1-V are internally shorted
Analog Output CH1	CH1+V	0	0	1	0	0	AL	VFD (GS1-20P2) though Ziplink ZL-RTB20	0–10 V speed command to VFD
Module Logic +24 V	+V1	0	0	0	0	1	VDC +24V	PLC Base (D2-09B-1) through ZipLink ZL-RTB20	Powers analog output circuitry, also fused with 1A fast blow fuse
Module Logic 0 V	0V1	0	0	0	0	1	VDC – 0V	PLC Base (D2-09B-1) through ZipLink ZL-RTB20	Common return for module logic
Slot Power Feed	Backplane	0	0	0	0	1	PLC Base Backplane	Internal slot power	No external wiring
Total Points		**0**	**0**	**2**	**0**	**3**			

Sheet: Input Module (D2-08ND3)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
D2-08ND3 X0	X0	0	1	0	0	0	Forward Pushbutton	Forward button S2PR-P3G with Contact block SA-CA with LED module SA-LDG	Shared button +24V -> NO contact -> X0.
D2-08ND3 X1	X1	0	1	0	0	0	Reverse Pushbutton	Reverse button S2PR-P3Y with Contact block SA-CA with LED module SA-LDY	Shared button +24V -> NO contact -> X1.
D2-08ND3 X2	X2	0	1	0	0	0	Run 1 / Deadman 1 Pushbutton	Run 1 button S2PR-P3B with Contact block SA-CA with LED module SA-LDB	Shared button +24V -> NO contact -> X2.
D2-08ND3 X3	X3	0	1	0	0	0	Run 2 / Deadman 2 Pushbutton	Run 2 button S2PR-P3B with Contact block SA-CA with LED module SA-LDB	Shared button +24V -> NO contact -> X3.
D2-08ND3 X4	X4	0	1	0	0	0	E-Stop Button Monitor	E-stop button E22B1 with NC contact block, wired in series with Mitsubishi SD-N35 contactor coil	PSU +V -> E22B1 NC -> SD-N35 A1 (X4 tapped between E22B1 and A1). SD-N35 A2 -> PSU -V.
D2-08ND3 X5	X5	0	1	0	0	0	VFD Fault Relay	VFD fault output Huanyang HY02D211B-T internal SPDT relay (FA/FC/FB), programmed as fault indication via PD052=02	PSU +V -> VFD FA. VFD FC -> X5. FB unused.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
D2-08ND3 X6	X6	0	1	0	0	0	Reserved - Reset Pushbutton (planned)	Reset button S2PR-P3R with Contact block SA-CA with LED module SA-LDR	Unwired. Planned: shared button +24V -> NO contact -> X6 (same topology as X0-X3).
D2-08ND3 X7	X7	0	1	0	0	0	Spare	Unused	Unwired.
D2-08ND3 Input Common (C)	C	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail. Both top-row and bottom-row C terminals on the D2-08ND3 are internally tied.	C -> PSU -V. Module wired in sourcing mode. Shared +24V button supply is fed from PSU +V through a 1 A fast-blow fuse (EURO S10H-5H holder) into the EURO S4LH distribution terminal, which provides the common +24V to X0/X1/X2/X3 (and X6 when wired).
Total Points		**0**	**8**	**0**	**0**	**1**			

Sheet: Analog Input (F2-08AD-1)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
F2-08AD-1 +24V (Module Power)	+24V	0	0	0	0	1	Mean Well NDR-480-24 +V	DC power supply +24V rail	PSU +V -> Module +24V terminal directly. No fuse on this branch.
F2-08AD-1 0V (Module Power Return)	0V	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail. Shared with VFD DCM and ACM (all tied to PSU -V).	PSU -V -> Module 0V terminal. Also serves as the common reference for channel inputs.
F2-08AD-1 CH1+ (Channel 1 Signal)	CH1+	0	0	0	1	0	VFD Motor Current Monitor	VFD analog output Huanyang HY02D211B-T VO terminal, configured for output current via PD054=1 (0-10V proportional to motor current)	VFD VO -> Module CH1+ terminal. Signal return reference via PSU -V (VFD ACM also tied to PSU -V).
F2-08AD-1 CH2	CH2	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH3	CH3	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH4	CH4	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH5	CH5	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH6	CH6	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH7	CH7	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH8	CH8	0	0	0	1	0	Spare	Unused channel	Unwired.
Total Points		**0**	**0**	**0**	**8**	**2**			

Sheet: RTD Input (F2-04RTD)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
F2-04RTD +24V (Module Power)	+24V	0	0	0	0	1	Mean Well NDR-480-24 +V	DC power supply +24V rail	PSU +V -> Module +24V terminal. Module installed in rack but no channels currently used in this build.
F2-04RTD 0V (Module Power Return)	0V	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail	PSU -V -> Module 0V terminal.
F2-04RTD CH1	CH1	0	0	0	1	0	Spare	Unused RTD input channel	Unwired. Available for future RTD temperature sensor.
F2-04RTD CH2	CH2	0	0	0	1	0	Spare	Unused RTD input channel	Unwired.
F2-04RTD CH3	CH3	0	0	0	1	0	Spare	Unused RTD input channel	Unwired.
F2-04RTD CH4	CH4	0	0	0	1	0	Spare	Unused RTD input channel	Unwired.
Total Points		**0**	**0**	**0**	**4**	**2**			

Sheet: Relay Output (D2-08TR)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
D2-08TR Y0	Y0	1	0	0	0	0	Forward Indicator LED	LED module SA-LDG (green, 12-24 VDC) integrated in Forward pushbutton S2PR-P3G (assembly also drives X0)	PSU +V -> SA-LDG (+). SA-LDG (-) -> Y0 contact. Relay sinks Y0 to -V when energized, lighting the LED.
D2-08TR Y1	Y1	1	0	0	0	0	Reverse Indicator LED	LED module SA-LDY (yellow, 12-24 VDC) integrated in Reverse pushbutton S2PR-P3Y (assembly also drives X1)	PSU +V -> SA-LDY (+). SA-LDY (-) -> Y1 contact. Relay sinks Y1 to -V when energized.
D2-08TR Y2	Y2	1	0	0	0	0	Run 1 / Deadman 1 Indicator LED	LED module SA-LDB (blue, 12-24 VDC) integrated in Run 1 pushbutton S2PR-P3B #1 (assembly also drives X2)	PSU +V -> SA-LDB (+). SA-LDB (-) -> Y2 contact.
D2-08TR Y3	Y3	1	0	0	0	0	Run 2 / Deadman 2 Indicator LED	LED module SA-LDB (blue, 12-24 VDC) integrated in Run 2 pushbutton S2PR-P3B #2 (assembly also drives X3)	PSU +V -> SA-LDB (+). SA-LDB (-) -> Y3 contact.
D2-08TR Y4	Y4	1	0	0	0	0	Error / Fault Indicator	Wamco model 525 panel-mount LED indicator (28 VDC, 0.6 W)	PSU +V -> Wamco 525 (+). Wamco 525 (-) -> Y4 contact. 28V-rated indicator slightly under-driven at 24V (acceptable, ~21 mA draw).

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
D2-08TR Y5	Y5	1	0	0	0	0	VFD Fault Reset	Huanyang HY02D211B-T VFD RST terminal (multi-input, sinking/NPN mode)	Y5 -> VFD RST. When Y5 energized, relay closes Y5 to common (PSU -V), pulling RST input to DCM. ~500 ms pulse during PLC unjam routine.
D2-08TR Y6	Y6	1	0	0	0	0	VFD Forward Run	Huanyang HY02D211B-T VFD FOR terminal (multi-input, sinking/NPN mode)	Y6 -> VFD FOR. When Y6 energized, pulls FOR to DCM. Forward shredding.
D2-08TR Y7	Y7	1	0	0	0	0	VFD Reverse Run	Huanyang HY02D211B-T VFD REV terminal (multi-input, sinking/NPN mode)	Y7 -> VFD REV. When Y7 energized, pulls REV to DCM. Used during PLC unjam routine.
D2-08TR Relay Common (C)	C	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail	C -> PSU -V. Sinking mode: when any Y0-Y7 closes, that terminal connects to -V.
Total Points		**8**	**0**	**0**	**0**	**1**			

Sheet: Analog Output (F2-08DA-2)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
F2-08DA-2 +24V (Module Power)	+24V	0	0	0	0	1	Mean Well NDR-480-24 +V	DC power supply +24V rail	PSU +V -> Module +24V terminal directly. No fuse on this branch.
F2-08DA-2 0V (Module Power Return / Analog Common)	0V	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail. Single 0V terminal serves both as power return and analog signal reference for all 8 channels.	PSU -V -> Module 0V terminal. VFD ACM is also tied to PSU -V, so the analog signal return is shared through the common 0V rail.
F2-08DA-2 +V1 (Channel 1 Signal)	+V1	0	0	1	0	0	VFD Analog Speed Reference	Huanyang HY02D211B-T VFD VI terminal (analog voltage input, 0-10 V), used as frequency/RPM reference. Requires PD002=1.	+V1 -> VFD VI. 0-10 V signal scaled to commanded frequency: 0 V = 0 Hz, 10 V = max frequency.
F2-08DA-2 +V2	+V2	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V3	+V3	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V4	+V4	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V5	+V5	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V6	+V6	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V7	+V7	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V8	+V8	0	0	1	0	0	Spare	Unused channel	Unwired.
Total Points		**0**	**0**	**8**	**0**	**2**			

Sheet: VFD (Huanyang HY02D211B-T)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD R (AC Line Hot Input)	R	0	0	0	0	1	AC Mains Hot via main disconnect and 30 A breaker	120 VAC line conductor from wall outlet, routed through panel main disconnect and a 30 A C-curve DIN-rail circuit breaker	Wall AC L -> disconnect -> 30 A breaker -> VFD R. Black wire per NFPA 79.
VFD T (AC Neutral Input via Contactor and Breaker)	T	0	0	0	0	1	AC Mains Neutral via Phoenix Contact TMC 8 3C 15A breaker and Mitsubishi SD-N35 contactor	Neutral leg switched through the safety-disconnect contactor. As-built path: Wall N -> Phoenix Contact TMC 8 3C 15A breaker (P/N 2907618) -> SD-N35 T1 -> contactor pole -> SD-N35 L1 -> VFD T.	AS-BUILT: neutral leg is switched by the contactor instead of the hot leg. Note: NFPA 79 / NEC standard practice is to switch the HOT leg. Review for final report. When E-stop is pressed, SD-N35 coil drops -> contact opens -> VFD T loses neutral -> VFD shuts down.
VFD Ground (PE)	Ground	0	0	0	0	1	Panel Ground Bus	Equipment grounding conductor; bonds VFD chassis to panel ground bus	VFD ground stud -> panel ground bus. Green or Green/Yellow wire.
VFD U (Motor Phase 1 Output)	U	0	0	0	0	1	Banana Socket (Blue) -> Motor T1	Panel-mount banana socket (blue, 4 mm) provides quick-disconnect to GE 5KE182BC205B motor T1	VFD U -> blue banana socket -> motor terminal.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD V (Motor Phase 2 Output)	V	0	0	0	0	1	Banana Socket (White) -> Motor T2	Panel-mount banana socket (white, 4 mm) provides quick-disconnect to GE 5KE182BC205B motor T2	VFD V -> white banana socket -> motor terminal.
VFD W (Motor Phase 3 Output)	W	0	0	0	0	1	Banana Socket (Red) -> Motor T3	Panel-mount banana socket (red, 4 mm) provides quick-disconnect to GE 5KE182BC205B motor T3	VFD W -> red banana socket -> motor terminal.
VFD P+ (Brake Resistor +)	P+	0	0	0	0	1	Brake Resistor ATO APCS-300R30-AD (+ terminal)	Dynamic braking resistor 300 W, 30 Ohms, aluminum-clad (ATO model APCS-300R30-AD)	VFD P+ -> brake resistor terminal 1.
VFD PR (Brake Resistor Return)	PR	0	0	0	0	1	Brake Resistor ATO APCS-300R30-AD (return terminal)	Dynamic braking resistor (same component as P+ row)	VFD PR -> brake resistor terminal 2.
VFD UPF	UPF	0	1	0	0	0	Unused	Multi-Output 2 (PD051), open-collector optocoupler	Unwired.
VFD DRV	DRV	0	1	0	0	0	Unused	Multi-Output 1 (PD050), open-collector optocoupler	Unwired.

Point Description	**Origin**	**DO**	**DI**	**AO**	**AI**	**Pwr**	**Destination Address**	**Destination Description**	**Notes**
VFD DCM (Digital Common, top row)	DCM	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return - shared reference for FOR/REV/RST sinking inputs	VFD DCM (top row) -> PSU -V. PSU -V also feeds D2-08TR relay common.
VFD SPL	SPL	0	1	0	0	0	Unused	Multi-Input 6 - default low-speed select	Unwired.
VFD SPM	SPM	0	1	0	0	0	Unused	Multi-Input 5 - default middle-speed select	Unwired.
VFD SPH	SPH	0	1	0	0	0	Unused	Multi-Input 4 - default high-speed select	Unwired.
VFD RST (Fault Reset Input)	RST	0	1	0	0	0	D2-08TR Y5	PLC relay output (sinking NPN mode) driving fault reset	PLC Y5 -> VFD RST. Pulled to DCM (0V) when Y5 closes.
VFD REV (Reverse Run Input)	REV	0	1	0	0	0	D2-08TR Y7	PLC relay output (sinking NPN mode) driving reverse run command	PLC Y7 -> VFD REV. Pulled to DCM (0V) when Y7 closes.
VFD FOR (Forward Run Input)	FOR	0	1	0	0	0	D2-08TR Y6	PLC relay output (sinking NPN mode) driving forward run command	PLC Y6 -> VFD FOR. Pulled to DCM (0V) when Y6 closes.
VFD ACM (Analog Common, top row)	ACM	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return - shared reference for analog signals	VFD ACM (top row) -> PSU -V. Shared 0V reference for VI (speed in) and VO (current monitor out).
VFD VO (Analog Output, 0-10 V)	VO	0	0	1	0	0	F2-08AD-1 CH1+	PLC analog input module CH1, used as motor current monitor (PD054=1)	VFD VO -> F2-08AD-1 CH1+. 0-10 V scaled to motor output current.
VFD 10V	10V	0	0	1	0	0	Unused	+10 V reference output (for external potentiometer)	Unwired.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD FA (Multi-Output 3 Relay NO)	FA	0	1	0	0	0	Mean Well NDR-480-24 +V	DC power supply +V rail	PSU +V -> VFD FA. No fuse on this branch. FA-FC closes when VFD trips on any fault (PD052=02).
VFD FC (Multi-Output 3 Relay Common)	FC	0	1	0	0	0	D2-08ND3 X5	PLC discrete input X5 (jam signal)	VFD FC -> PLC X5. When FA-FC closes, +24V from FA reaches X5 (PLC reads HIGH = jam).
VFD FB (Multi-Output 3 Relay NC)	FB	0	1	0	0	0	Unused	Normally-closed contact - not used in active-high NO configuration	Unwired.
VFD 24V (Onboard 24V Supply)	24V	0	0	0	0	1	Unused	Onboard 24 VDC auxiliary supply, max 200 mA	Unwired. Panel uses external Mean Well NDR-480-24 instead.
VFD DCM (bottom row)	DCM (bottom)	0	0	0	0	1	(same node as top-row DCM)	Internally tied to top-row DCM	Alternate landing point. Top-row DCM is used.
VFD 5V	5V	0	0	1	0	0	Unused	+5 V reference output	Unwired.
VFD ACM (bottom row)	ACM (bottom)	0	0	0	0	1	(same node as top-row ACM)	Internally tied to top-row ACM	Alternate landing point. Top-row ACM is used.
VFD AI (Analog Current Input, 4-20 mA)	AI	0	0	0	1	0	Unused	Analog current input for speed reference. Speed command uses VI voltage input instead.	Unwired.
VFD VI (Analog Voltage Input, 0-10 V)	VI	0	0	0	1	0	F2-08DA-2 +V1	PLC analog output module CH1, used as commanded frequency reference (PD002=1)	F2-08DA-2 +V1 -> VFD VI. 0-10 V scaled to commanded frequency.
VFD RS-	RS-	0	0	0	0	1	Unused	RS-485 Modbus communication negative line	Unwired. Modbus not used.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD RS+	RS+	0	0	0	0	1	Unused	RS-485 Modbus communication positive line	Unwired.
Total Points		**0**	**11**	**3**	**2**	**15**			

16.3 Appendix C: PLC Ladder Logic — Reference

The complete 31-rung Do-more Designer ladder export is reproduced inline in Section 6.1 (PLC Ladder Logic Architecture), with each page of `electrical/Circuit_Logic.pdf` placed adjacent to the rung-by-rung walkthrough that describes its contents. The full PDF source is also retained at `electrical/Circuit_Logic.pdf` in the repository for archival and detailed inspection.

16.4 Appendix D: VFD Parameter Settings

The full programmed parameter set — code, function, value, and rationale — is given in Table 1 (Section 5.2, Motor and VFD). Motor values `PD141`–`PD144` as currently programmed reflect the GE 5KE182BC205B bench motor (230 V, 7.6 A, 4 poles, 1750 RPM); the as-built production motor is the IronHorse MTCP2-002 (230 V, 5.93 A, 4 poles, 1735 RPM), and re-programming `PD141`–`PD144` to the IronHorse nameplate plus re-checking the `PD124` over-torque trip (150 % of `PD142` → 8.9 A on the IronHorse FLA, versus 11.4 A on the bench value) is captured in the open-work punch list (`docs/term2_week11_Open_Work.md`, item 4).

Power / braking wiring: R = AC hot through the SD-N35 contactor (and the 30 A branch breaker); T = AC neutral; U/V/W to motor via colored banana sockets; P+/PR to a 300 W, 30 Ω brake resistor (ATO APCS-300R30-AD) to absorb regenerative energy during reversal in the jam-clear cycle. Control wiring is summarized in Appendix B.

A firmware-specific finding worth carrying forward: `PD052 = 12` (continue-running over-torque indication) did not fire the FA/FC relay on the Huanyang revision used in this build. The working configuration is `PD052 = 02` with `PD123 = 3` — over-torque self-trips, which fires the fault relay, which is read by the PLC on X5.

16.5 Appendix E: Full PLC Variable Reference

The compact key-terms table in Section 6.1 (PLC Ladder Logic Architecture) gives a one-phrase meaning for every tag used in the ladder. The table below is the full address-by-address reference: every X / Y / WX / WY / C / T / CT / V / ST point used in the program, with its electrical type or instruction kind and a longer description that names the source module, the ladder context, and any related rungs.

Address	Name in Ladder	Type	Description
X0	FORWARDS	DI	Forward pushbutton (green S2PR-P3G). HIGH = pressed.

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Address	Name in Ladder	Type	Description
X1	REVERSE	DI	Reverse pushbutton (yellow S2PR-P3Y). HIGH = pressed.
X2	DEADMAN_A_OK	DI	Deadman button A (blue #1). Must be held for DEADMEN_OK.
X3	DEADMAN_B_OK	DI	Deadman button B (blue #2). Must be held simultaneously with X2.
X4	STOP_PB	DI	E-stop NC contact monitor. LOW when E-stop is pressed. Used as NOT X4 in rungs 17, 18.
X5	OVERCURRENT	DI	VFD fault relay input (FA/FC, PD052=02). HIGH = VFD tripped. Triggers jam recovery sequence.
X7	CAP_TOUCH_TRIP	DI	Capacitive touch safety sensor (on-hand, wired). HIGH = safety zone breached. Immediately sets STOP_MOTOR, CTSI_TRIG.
Y0	FWD_LED	DO	Forward indicator LED (green SA-LDG). ON when FORWARDS_SET (C8) is HIGH.
Y1	REVERSE_LED	DO	Reverse indicator LED (yellow SA-LDY). ON when REVERSE_SET (C9) is HIGH.
Y2	DMAN_A_LED	DO	Deadman A LED (blue #1). ON when (X2 OR C23) AND NOT Y4.
Y3	DMAN_B_LED	DO	Deadman B LED (blue #2). ON when (X3 OR C24) AND NOT Y4.
Y4	E_LIGHT	DO	Error / fault indicator (Wamco 525 panel LED). SET by E-stop open, CAP_TOUCH, and CT0.Done. RST by ERROR_RESET (C18).

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Address	Name in Ladder	Type	Description
Y5	VFD_RESET	DO	VFD fault reset output (RST terminal). Energized while DEADMEN_OK AND X5 are both HIGH (rung 19).
Y6	VFD_RUN_FWD	DO	VFD forward run (FOR terminal). Energized by rung 9 when all permissives are met and no fault/pause active.
Y7	VFD_RUN_RVS	DO	VFD reverse run (REV terminal). Energized by rung 10 via REVERSE_SET or unconditionally by UNJAM_R (C13).
WX0	Analog input word	AI word	Raw ADC from F2-08AD-1 CH1 (VFD VO, motor current). Input to rung 3 SCALE (0–4095 → 0–140).
WY0	Analog output word	AO word	Raw DAC to F2-08DA-2 CH1 (VFD VI, speed reference). Output of rung 2 SCALE (V2000 0–1600 → 0–4095).
C0	DEADMEN_OK	BOOL	HIGH when (X2 OR C23) AND (X3 OR C24). Both deadman paths held.
C1	HMI_STP_REQ	BOOL	HMI STOP button request. Rung 7: NOT C1 gates RUN_REQUEST_SET.
C3	PERMISSIVE_OK	BOOL	Mirrors DEADMEN_OK (C0). Used as interlock in rungs 7 and 8.
C8	FORWARDS_SET	BOOL latch	Forward direction selected. SET by X0 (rung 13) or on first scan init (rung 4). RST by X1.

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Address	Name in Ladder	Type	Description
C9	REVERSE_SET	BOOL latch	Reverse direction selected. SET by X1 (rung 14). RST by X0. RSTR clears C9–C30 on first scan.
C10	JAM_LATCH	BOOL	Output of T0 (OFFDTMR). HIGH while overcurrent is active; stays HIGH 2 s after X5 drops. Blocks Y6, drives T1.
C11	init flag	BOOL	SET on first scan by $FirstScan (rung 1 RST, rung 4 triggers init). Used to initialize FORWARDS_SET and clear C9–C30.
C12	RUN_REQUEST	BOOL latch	Motor run command. SET by rung 7, RST by rung 8 (NOT X4 or NOT PERMISSIVE_OK). Also shown as MOTOR_STATUS on HMI.
C13	UNJAM_R	BOOL	Output of T1 (OFFDTMR, 2 s). HIGH while JAM_LATCH is HIGH; stays HIGH 2 s after JAM_LATCH drops. Energizes Y7 (reverse) unconditionally.
C15	STOP_MOTOR	BOOL latch	Hard stop flag. SET by CAP_TOUCH or CT0.Done (lockout). RST by ERROR_RESET (C18). Blocks Y6 and Y7 in rungs 9, 10.
C16	LOCKOUT_TMR_STRT	BOOL latch	Enables T2 accumulating timer. SET when UNJAM_R active and T2 not timing. RST when T2.DN fires (30 s accumulated).
C17	PAUSE_MOTOR	BOOL	Output of T3 (OFFDTMR, 2 s). Triggered by T1.DN rising edge. Blocks Y6 for 2 s after reverse completes (coast-to-stop buffer).

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Address	Name in Ladder	Type	Description
C18	ERROR_RESET	BOOL	Fault reset coil (from HMI ERROR_OFF button, Screen 3). RSTs STOP_MOTOR, E_LIGHT, JAMMED_MOTOR, CTSI_TRIG, CHANGE_SCREEN, CT0.
C20	JAMMED_MOTOR	BOOL latch	Lockout flag. SET when CT0.Done fires (3rd jam). Shown on Screen 3 SHREDDER JAM indicator. RST by ERROR_RESET.
C21	CTSI_TRIG	BOOL latch	Capacitive touch triggered. SET by X7 (rung 28) or ERROR_RESET path. Shown on Screen 3. RST by ERROR_RESET.
C22	CHANGE_SCREEN	BOOL latch	Triggers HMI screen change to error screen (Screen 3). Rung 29: sets V100=3 and starts T4 (3 s). T4.DN resets V100 to 0.
C23	TESTMAN_A_OK	BOOL	HMI soft deadman A (from Test Screen PUSHBUTTON A toggle, D-MAN A tag). ORed with X2 in rung 5.
C24	TESTMAN_B_OK	BOOL	HMI soft deadman B (from Test Screen PUSHBUTTON B toggle, D-MAN B tag). ORed with X3 in rung 5.
C25	OVERCURRENT_TEST	BOOL	HMI soft overcurrent trigger (Test Screen OVERCURRENT momentary button). ORed with X5 in rung 19.
C26	E_STOP_TRIG	BOOL	E-stop triggered flag. SET by rung 17 when NOT X4. Shown on Screen 3 EMERGENCY STOP indicator.

continued on next page . . .

Address	Name in Ladder	Type	Description
C99	T4 output	BOOL	Output of T4 off-delay timer (3 s). Active 3 s after CHANGE_SCREEN sets. Not used further in current build.
T0	Overcurrent off-delay	OFFDTMR 2 s	Input: DEADMEN_OK AND (X5 OR C25). Output: JAM_LATCH (C10). Extends jam latch 2 s after overcurrent clears.
T1	Unjam reverse off-delay	OFFDTMR 2 s	Input: JAM_LATCH (C10). Output: UNJAM_R (C13). Drives reverse for 2 s after JAM_LATCH drops.
T2	Lockout accumulating timer	TMRA 30 s	Accumulates total time LOCKOUT_TMR_STRT (C16) is active. T2.DN at 30 s resets C16, stops counting new jams.
T3	Post-reverse pause	OFFDTMR 2 s	Triggered by T1.DN rising edge. Output: PAUSE_MOTOR (C17). Holds Y6 LOW for 2 s after reverse finishes.
T4	Screen-change timeout	OFFDTMR 3 s	Input: CHANGE_SCREEN (C22). After 3 s, T4.DN fires rung 30: resets V100 to 0 (HMI returns from error screen).
CT0	JAM_DETECT	CTU preset 3	Counts jam events. UP input: JAM_LATCH AND NOT T2.DN. RST input: ERROR_RESET (C18). CT0.Done (N0=3) triggers lockout.
V100	ERROR_SCREEN	Int (BCD)	HMI screen selector written by ladder. Rung 29 writes 3 (→ Screen 3), rung 30 writes 0 after T4 (3 s).

continued on next page . . .

Address	Name in Ladder	Type	Description
V2000	RPM_CTRL	Int 16	Speed setpoint (0–1600 counts). Source for rung 2 SCALE → WY0. Written by HMI IncDec buttons.
V3000	CURRENT	Int 16	Scaled motor current (0–140). Output of rung 3 SCALE from WX0. Displayed on Screen 2 CURRENT field.
ST0	$FirstScan	System bit	Do-more system coil HIGH for exactly one scan on power-up. Used in rung 1 to RST C11 (init trigger).
ST1	$On	System bit	Do-more system coil always HIGH. Used in rungs 2 and 3 to run SCALE instructions every scan.

16.6 Appendix G: Test Plan

Test Plan v1.0 comprises a top-down power-on walkthrough followed by ten bottom-up test cases. Each case is exercised against the must/should requirements (M-series).

ID	Test case
Top-down	Power-on walkthrough: AC in → PSU → PLC boot → HMI IDLE/READY
BU-01	AC power input and fuse distribution
BU-02	24 VDC PSU output (Mean Well NDR-480-24)
BU-03	PLC rack initialization (DirectLOGIC 205)
BU-04	E-stop hardware safety circuit
BU-05	Two-hand start (deadman) safety logic
BU-06	Lid interlock (n/a in this build — access guarded mechanically by the ME team)
BU-07	VFD speed command and motor RPM (HY02D211B-T)
BU-08	Overcurrent / overload protection
BU-09	HMI operator interface (EA1-T4CL C-More Micro-Graphic)
BU-10	Status indicator lights (power / running / fault)

Validation results to date are summarized in the Testing and Validation and Results sections; load testing under real shredding load is pending mechanical integration.

17 Authors and Acknowledgements

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Disclaimer

This report documents the as-built state of the Universal Plastic Shredder V2.0 electrical system at end of the ECE 412 / 413 senior capstone term. Findings, parameters, and as-built details reflect the validated bench configuration; system-level validation against the ME team's mechanical assembly is deferred to future work as captured on the open-work handover punch list (`docs/term2_week11_Open_Work.md`).

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2. Approval Statement

By signing below, the industry sponsor certifies that they have reviewed this final report (Revision 1.0) and approve it as the record of work completed by the Electrical Engineering team on the Universal Plastic Shredder V2.0 senior capstone project. The signature indicates technical concurrence with the as-built design, acknowledgement of the open items captured on the handover punch list (`docs/term2_week11_Open_Work.md`), and authorization to release this document to the sponsor and the institutional archive. This document is provided "as built"; any subsequent revisions shall be issued under a new revision number and require renewed approval.

3. Sponsor Approval

Role	Name & Affiliation	Signature	Date
Industry Sponsor	Bridget Lannigan MME / Ichor Systems	____________________	____________